

Optimal Consumption and Investment Strategies with a Perishable and an Indivisible Durable Consumption Good*

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Abstract. We study the consumption and investment choice of an agent in a continuous-time economy with a riskless asset, several risky financial assets, and two consumption goods, namely a perishable and a durable good with an uncertain price evolution. Assuming lognormal prices and a multiplicatively separable, isoelastic utility function, we provide an explicit Merton-type solution for the optimal strategies for the case where the durable (and all other assets) can be traded without transaction costs. For the case where the durable good is indivisible, in the sense that durable trades imply transaction costs proportional to the value of the current durable holdings, we show analytically that the optimal durable trading policy is characterized by three constants $\underline{z} < z^* < \bar{z}$. As long as the ratio z of the total current wealth to the value of current durable holdings of the investor is in $(\underline{z}, \bar{z})$, it is optimal not to trade the durable. At the boundaries of this interval it is optimal to trade the durable to attain $z = z^*$. The model is used to examine the optimal substitution between perishable and durable consumption and the importance of the durable price uncertainty and the correlation between the price of the durable good and financial asset prices.

Keywords. Optimal consumption and investment, durable goods, transaction costs, stochastic control, optimal stopping, viscosity solutions, numerical solution.

JEL classification numbers. G11, D11, D91, C61.

1 Introduction

Modern economies offer an enormous variety of consumption goods. For modeling purposes each good is typically classified either as a *perishable* good that cannot be stored and only provides utility at the time of purchase or as a *durable* good that provides utility to its owner over a period of time and can be resold so that it also acts as an investment that transfers wealth over time. Traditional models of optimal consumption and investment problems consider either a single perishable consumption good, cf. Merton (1969), or a single durable consumption good, cf. Grossman and Laroque (1990). In this paper, we merge these settings by allowing for both a perishable and a durable good with a stochastically evolving relative price. This enables us to study optimal behavior in an economically more appealing set-up and to address questions that cannot be dealt with in traditional single-good models, such as how optimal perishable and durable consumption policies are related and how the uncertainty about future relative consumption prices affects optimal consumption and investment decisions.

More specifically, we examine the optimal consumption and investment choice of an agent in a continuous-time economy with one riskless and several risky financial assets and both a perishable and a durable consumption good. The durable consumption good is indivisible in the sense that in order to change the stock of the good (beyond the assumed depreciation), the agent must sell his entire current holdings of the good and then buy the desired new stock, which is the case for houses and cars for example. We assume that in doing so the agent must pay transaction costs proportional to the value of the current stock of the durable. The perishable good and the financial assets are traded without transaction costs. The agent extracts utility from the rate of consumption of the perishable good and the stock of the durable good. We study the case where the agent has an infinite time horizon and a utility function of the multiplicatively separable, isoelastic form $U(c, k) = (c^\beta k^{1-\beta})^{1-\gamma} / (1 - \gamma)$, where c and k are the current perishable consumption rate and the current stock of the durable, respectively. Furthermore, measured in terms of perishable consumption units, the price of the durable good and the prices of the risky financial assets follow correlated geometric Brownian motions.

Our first contribution is to derive an explicit solution to the consumption-investment problem for the case of no transaction costs. The optimal strategy is to keep both the perishable consumption rate, the value of the durable holdings, and the amount invested in each of the risky financial assets as fixed fractions of wealth. This result generalizes the solution to the single perishable consumption good problem of Merton (1969). The set of risky financial assets exhibits two-fund separation in that the optimal investment strategy combines only the mean-variance tangency portfolio and a durable hedge portfolio, which is the portfolio with the highest possible absolute correlation with the price of the durable consumption good.

The optimal strategy for the no transaction costs problem involves continuous rebalancing of the stock of the durable due to fluctuations in financial asset prices and the price of the durable and also due to the physical depreciation of the stock of the durable. With transaction costs, such a strategy is clearly not optimal. Our second contribution is to characterize the optimal consumption and investment

policies with transaction costs. We show that the optimal behavior is completely determined by the ratio $z = x/(kp)$ of the total current wealth, x , to the product of the current stock of durable, k , and the current price of durable, p . In addition, using the notion of viscosity solutions, we demonstrate that there are two critical values $\underline{z} < \bar{z}$ such that it is optimal to refrain from trading the durable good as long as $z \in (\underline{z}, \bar{z})$. At the boundaries of this interval, it is optimal to trade the durable. When $z = \underline{z}$, it is optimal to shift to a lower stock of the durable, and when $z = \bar{z}$, it is optimal to shift to a higher stock of the durable. In both cases, the optimal transaction is such that, immediately after the transaction, the new value of z , z^* , is in the open interval $(\underline{z}, \bar{z})$. If the initial values are such that $z \notin (\underline{z}, \bar{z})$, the optimal policy involves an initial transaction to $z^* \in (\underline{z}, \bar{z})$. Since there is a rather small loss in utility from deviating a little from the optimal durable consumption level, adjustments of the durable holdings will be infrequent even with small transaction costs. Concerning the optimal investment strategies, we show that the presence of transaction costs on the durable good does not change the fund separating structure of the set of risky financial assets.

Our third contribution is to provide numerical results illustrating several important economic effects of transaction costs in our setting. We find that the relative risk aversion associated with the indirect utility of wealth varies over the interval $(\underline{z}, \bar{z})$ such that the relative risk aversion is low close to the boundaries (and, in particular, immediately before a durable trade) and high near the target value z^* (immediately after a durable trade). We show that this variation in risk aversion has a significant impact on the weights of the two separating risky funds in the optimal portfolio. We demonstrate that the no-trade interval $(\underline{z}, \bar{z})$ widens as transaction costs increase and that transaction costs generally have smaller quantitative effects than in a model with only a durable consumption good (the Grossman and Laroque model discussed below), since the agent in our model will substitute perishable consumption for durable consumption to reduce transaction costs. For example, the expected period of time between durable trades is significantly lower than found by Grossman and Laroque. We find that the perishable consumption propensity, i.e. the optimal rate of perishable consumption as a fraction of total wealth, can vary substantially over the interval $(\underline{z}, \bar{z})$, in contrast to the no transaction costs case where it is kept constant. The precise relation is highly dependent on the exogenous preference parameters β and γ and the endogenously determined relative risk aversion associated with the indirect utility of wealth. For some parameter values, the perishable consumption propensity is a decreasing function of z and, hence, increasing in the price of the durable and decreasing in financial wealth. For other parameter values, the relation is non-monotonic. Finally, we show that the optimal behavior can be highly sensitive to the correlation between the price of the durable good and financial asset prices and, especially, the risk-return relationship of the durable good.

The analysis of the paper generalizes that of Grossman and Laroque (1990) who consider the simpler problem with only a durable consumption good that acts as the numeraire good. In their simpler set-up, they give a similar characterization of the optimal durable trading strategy and have similar results on risk aversion and the dependence of the no-trade region on transaction costs. Our analysis shows that their approach and conclusions carry over to the more general and economically more appealing framework

with two types of consumption goods with an uncertain relative price, but that the quantitative effects of transaction costs are smaller in the two-good economy. Moreover, our general setting allows us to study the relation between perishable and durable consumption and the impact of the uncertainty of the durable good price and its correlation with financial asset prices on optimal behavior.

A few other papers study the implications of durable consumption goods on optimal behavior. Cuoco and Liu (2000) examine a model with a single consumption good, namely a *divisible* durable good, e.g. furniture, where adjustment of the stock of the durable requires the payment of transaction costs proportional to the *change* in the stock of durable, not the current stock. Under this assumption it is optimal to keep the ratio z in a closed interval $[z_l, z_u]$, but at the boundaries the optimal transaction is the minimal needed to keep z in this interval, an effect also found in models with proportional costs of transacting financial assets, cf., e.g., Davis and Norman (1990). In mathematical terms, the optimal durable trading strategy is of a local time nature. Hence, the adjustments to the stock of the durable are small and frequent (continuous at the boundaries), contrary to our set-up where the changes are infrequent, non-infinitesimal jumps into the interior of the interval. The reason for this difference is that in our model the transaction costs are similar to a fixed cost in an optimal stopping problem.

Hindy and Huang (1993) discuss a model with utility derived from the stock of a durable good that cannot be resold once bought. With power utility and lognormal financial asset prices, it is optimal to keep the ratio of wealth to the stock of durable below a critical level and purchase durable good only at this critical level.

Detemple and Giannikos (1996) study a model with both a perishable and a durable good where the durable good provides a utility both through current purchases of the good (“status”) and through the current stock of the good (“services”). The investor is not allowed to sell out of his stock of durable and the price of the durable good (in terms of units of the perishable good) is spanned by the financial asset prices, whereas we allow for a durable-specific price risk. Using martingale methods for this complete markets optimization problem, they characterize the optimal consumption processes in terms of the state-price density.

The rest of the paper is organized as follows. Section 2 describes the details of the model we use. In Section 3, we provide an explicit solution to the utility maximization problem for the special case where the durable consumption good can be traded without transaction costs. For the problem with transaction costs, we derive analytically some important properties of the value function and the optimal strategies in Section 4. Since it seems impossible to derive explicit expressions in the latter case, we turn to numerical solution methods. Section 5 presents and discusses numerical results. Finally, Section 6 concludes the paper with a summary and a discussion of possible extensions. All proofs are in the appendices.

2 The model

We consider an economy with two consumption goods, namely a perishable good and a durable good, and $n+1$ financial securities. The price processes are defined below using the perishable good as the numeraire,

i.e. security prices and the price of the durable good are stated in terms of perishable consumption units. One of the financial assets is a riskfree security paying a constant continuously compounded interest rate $r \in \mathbb{R}_+$. The remaining n financial assets are risky. We model the uncertain evolution of the risky asset prices and the durable good price in terms of an n -dimensional Wiener process w_1 and a one-dimensional Wiener process w_2 uncorrelated with w_1 , where w_1 and w_2 are defined on a given probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The information structure of the economy is the filtration $\{\mathcal{F}_t\}_{t \geq 0}$ defined as the augmentation of the natural filtration generated by the Wiener process $w = (w_1, w_2)$. We assume that the n -dimensional price vector S_t of the risky financial assets evolves according to the stochastic differential equation

$$(2.1) \quad dS_t = \text{diag}(S_t) [\mu dt + \sigma dw_{1t}], \quad t \geq 0, \quad S_0 = s,$$

where σ is a constant, non-singular $n \times n$ matrix, μ is a constant vector in $\mathbb{R}^n$, and the term $\text{diag}(S_t)$ denotes the square matrix with the vector S_t along the diagonal and zero in all other entries. Hence, the prices of the risky financial assets are assumed to follow correlated geometric Brownian motions.

The unit price of the durable good, P_t , is also assumed to follow a geometric Brownian motion

$$(2.2) \quad dP_t = P_t [\mu_P dt + \sigma_{P1}^\top dw_{1t} + \sigma_{P2} dw_{2t}], \quad t \geq 0, \quad P_0 = p,$$

where μ_P and $\sigma_{P2} \neq 0$ are constant scalars, and σ_{P1} is a constant vector in $\mathbb{R}^n$. For future reference we define the $(n+1)$ -dimensional vector $\sigma_P = (\sigma_{P1}^\top, \sigma_{P2})^\top$. By construction, the price of the durable consumption good is imperfectly correlated with the financial asset prices, so that it is impossible to hedge away all the risk associated with the durable good price by appropriate investments in the financial market.

We assume that the stock of the durable good depreciates at a physical depreciation rate δ over time. More precisely, letting K_t be the number of units of the durable good held by an agent at time t , then, in time intervals where the agent does not trade the durable, K_t evolves according to

$$(2.3) \quad dK_t = -\delta K_t dt.$$

Moreover, we assume that it is costly to trade the durable good, i.e. to change the stock of the durable good beyond the depreciation. More precisely, we shall assume, following Grossman and Laroque (1990), that the transaction costs are a fraction $\lambda \in [0, 1]$ of the value $K_t P_t$ of the current stock of the durable. This is a reasonable assumption for *indivisible* durable goods, such as a house, where the only way one can change the position in the durable is to sell the current house and buy another house of the desired size and quality.¹ Cuoco and Liu (2000) consider *divisible* durable goods, where the transaction costs are proportional to the *change* in the durable stock, which is more reasonable for furniture, clothing, and similar durable consumption goods.

¹Our assumption models proportional transactions costs of selling and no transactions costs of buying durables, such that the price P_t is best interpreted as the unit purchase price of the durable. The model could easily be accommodated to proportional buying costs, but that would only complicate the notation and not change the qualitative results of the paper.

The agent must choose a strategy for consuming the perishable good and trading the durable good and the financial assets. Denote by C_t the agent's consumption rate of the perishable good at time t . A consumption strategy is a non-negative progressively measurable process, $C = \{C_t\}_{t \geq 0} \in \mathcal{L}^1$, where

$$\mathcal{L}^q = \left\{ \{\mathcal{F}_t\}\text{-adapted processes } X : \int_0^T \|X_u(\omega)\|^q du < \infty, \text{ for } P\text{-a.e. } \omega \in \Omega, \forall T > 0 \right\}, \quad q = 1, 2.$$

The set of consumption processes is denoted by $\mathcal{C}$. Again using the perishable consumption good as numeraire, we denote by Θ_{0t} the amount held in the riskfree security at time t . Similarly, for each $i \in \{1, \dots, n\}$, we let Θ_{it} denote the amount held in the i 'th risky security at time t , and define $\Theta_t = (\Theta_{1t}, \dots, \Theta_{nt})^\top$. As above, K_t denotes the holdings of the durable at time t , and the process $K = \{K_t\}_{t \geq 0}$ indirectly represents the trading strategy for the durable good. The set Π of trading strategies consists of the $(n+2)$ -dimensional progressively measurable stochastic processes (Θ_0, Θ, K) valued in $\mathbb{R} \times \mathbb{R}^n \times \mathbb{R}_+$ that satisfy $\Theta_0 \in \mathcal{L}^1$, $\Theta \in \mathcal{L}^2$, and $K \in \mathcal{L}^2$.

We define the wealth of the agent as the sum of his investments in the financial assets and the value of his current stock of the durable measured as units times the current unit purchase price. His time t wealth X_t is therefore given as

$$(2.4) \quad X_t = \Theta_{0t} + \Theta_t^\top \mathbf{1} + K_t P_t.$$

If the agent follows a perishable consumption strategy C and a self-financing trading strategy $\{\Theta_0, \Theta, K\}$, his wealth $X_t = X_t^{\Theta, K, C}$ evolves as

$$(2.5) \quad \begin{aligned} dX_t^{\Theta, K, C} = & \left[r(X_t^{\Theta, K, C} - K_t P_t) + \Theta_t^\top (\mu - r\mathbf{1}) + (\mu_p - \delta) K_t P_t - C_t \right] dt \\ & + [\Theta_t^\top \sigma + K_t P_t \sigma_{P1}^\top] dw_{1t} + K_t P_t \sigma_{P2} dw_{2t} \end{aligned}$$

in time intervals where the durable is not traded. At a time τ where the investor chooses to sell his durable good, there is a jump in his wealth due to the transaction costs:

$$(2.6) \quad X_\tau^{\Theta, K, C} = X_{\tau-}^{\Theta, K, C} - \lambda K_{\tau-} P_\tau.$$

We require that the agent chooses consumption and trading strategies satisfying the solvency condition that his total wealth is always greater than the transaction costs involved in an immediate sale of his durable holdings. To be more precise, assuming that the agent is endowed with $k = K_{0-}$ units of the durable good and a total wealth of $x = X_{0-}$, we say that a policy (Θ, K, C) is admissible if $(\Theta_0, \Theta, K) \in \Pi$, $C \in \mathcal{C}$, and

$$(2.7) \quad (X_t^{\Theta, K, C}, K_t, P_t) \in \bar{\mathcal{S}}, \quad P\text{-a.s. for all } t \geq 0,$$

where the solvency region $\bar{\mathcal{S}}$ is the closure of the set

$$\mathcal{S} = \{(x, k, p) \in \mathbb{R}_+^3 : x > \lambda k p\}.$$

We let $\mathcal{A}(x, k, p)$ denote the set of admissible policies. We shall assume that $(x, k, p) \in \bar{\mathcal{S}}$, which implies that $\mathcal{A}(x, k, p)$ is a non-empty set.

We consider an investor with an infinite time horizon and an additively time-separable utility function

$$\int_0^\infty e^{-\rho t} U(C_t, K_t) dt,$$

where $\rho > 0$ is a parameter reflecting the investor's time preferences. In the following we assume that the instantaneous utility function is of the multiplicatively separable, isoelastic form

$$(2.8) \quad U(c, k) = \frac{1}{1-\gamma} (c^\beta k^{1-\beta})^{1-\gamma},$$

where $\beta, \gamma \in (0, 1)$. A similar utility function was considered in a discrete-time setting with no transaction costs on the durable good by Dunn and Singleton (1986) and Lax (1999).^{2,3}

If the investor pursues the admissible policy $(\Theta, K, C) \in \mathcal{A}(x, k, p)$, his time zero expected utility is

$$J^{\Theta, K, C}(x, k, p) = \mathbb{E} \left[\int_0^\infty e^{-\rho t} U(C_t, K_t) dt \right].$$

Given an initial endowment such that $(X_{0-}, K_{0-}, P_0) = (x, k, p) \in \bar{\mathcal{S}}$, the investor's objective is to find the $(\Theta, K, C) \in \mathcal{A}(x, k, p)$ that maximizes his expected utility. Therefore, the investor's value function is defined as

$$(2.9) \quad V(x, k, p) = \sup_{(\Theta, K, C) \in \mathcal{A}(x, k, p)} J^{\Theta, K, C}(x, k, p).$$

3 An explicit solution to the no transaction costs problem

In this section we provide explicit expressions for the optimal consumption and investment strategies and the value function of the investor for the case where the position in the durable good can be changed costlessly corresponding to $\lambda = 0$. Our result generalizes the solution to the consumption-investment problem in the standard Merton setting, since we allow for both a perishable and a durable consumption good with a stochastically evolving relative price. The solution to the no transaction costs case also provides an important benchmark against which the optimal strategies with transaction costs will be compared and gives an upper bound on the value function with transaction costs.

When the stock of durable can be changed without costs, the consumption and investment policies can be chosen without regarding how total wealth is decomposed into financial wealth and the value of the durable stock. Therefore, the optimal policies and the value function will depend only on total current wealth x and the current price of durable p . We will denote the value function for the no transaction costs case by $\bar{V}(x, p)$ and the corresponding optimal policies by $\bar{C}$, $\bar{K}$, and $\bar{\Theta}$. In Appendix A

²It seems more reasonable that the instantaneous utility the agent gets from his durable holdings is related to the services he can extract from his stock of durable rather than the stock itself as we have assumed in (2.8). But if the services are proportional to the holdings of the durable and the utility of perishable consumption and durable services have the same form as in (2.8), our representation of preferences lead to the same optimal strategies.

³In their model with a durable good only, Grossman and Laroque (1990) assume a utility function of the form $U(K) = K^{1-\gamma}/(1-\gamma)$. Since they use the durable good as the numeraire, K in their model denotes the value of the durable holdings. Hence, they consider direct utility from the value of the durable, whereas we model direct utility from the physical stock of the durable, which seems more reasonable.

we derive explicit expressions for both the value function and the optimal policies. We do that by solving the Hamilton-Jacobi-Bellman (HJB) equation associated with $\bar{V}(x, p)$ and verifying the appropriate transversality condition.

Before we state the solution, we introduce some auxiliary parameters. Define the constants

$$\begin{aligned}\Lambda_0 &= -\frac{\rho}{\gamma} + \frac{1-\gamma}{\gamma} \left\{ r - (1-\beta)\mu_P + \frac{1}{2}(1-\beta)[1 + (1-\beta)(1-\gamma)] \|\sigma_P\|^2 \right\} \\ &\quad + \frac{1-\gamma}{2\gamma^2} (\mu - r\mathbf{1} - (1-\gamma)(1-\beta)\sigma\sigma_{P1})^\top (\sigma\sigma^\top)^{-1} (\mu - r\mathbf{1} - (1-\gamma)(1-\beta)\sigma\sigma_{P1}), \\ \Lambda_1 &= (1-\gamma)\sigma_{P2}^2 + \frac{1}{1-\beta} (r - \mu_P + \delta + (\mu - r\mathbf{1})^\top (\sigma^\top)^{-1} \sigma_{P1}), \\ \Lambda_2 &= \left(\frac{\gamma}{1-\beta} + \frac{1-\gamma}{2} \right) \sigma_{P2}^2.\end{aligned}$$

If $\Lambda_0 < 0$, the quadratic equation

$$(3.1) \quad \Lambda_0 + \Lambda_1 \alpha_k + \Lambda_2 \alpha_k^2 = 0$$

will have a single strictly positive solution, which we will denote α_k in the following. If we furthermore assume that

$$(3.2) \quad \Lambda_0 < -\frac{1}{2}(1-\gamma)\sigma_{P2}^2\alpha_k^2,$$

then the constant

$$\alpha_c = -\beta\Lambda_0 - \frac{1}{2}\beta(1-\gamma)\sigma_{P2}^2\alpha_k^2$$

will be positive, as will the constant

$$(3.3) \quad \alpha_v = \beta\alpha_c^{\beta(1-\gamma)-1}\alpha_k^{(1-\beta)(1-\gamma)}.$$

Finally, define the constant vector in $\mathbb{R}^n$

$$(3.4) \quad \alpha_\theta = \frac{1}{\gamma} (\sigma\sigma^\top)^{-1} (\mu - r\mathbf{1}) - \left(\frac{1-\beta(1-\gamma)}{\gamma} + \alpha_k - 1 \right) (\sigma^\top)^{-1} \sigma_{P1}.$$

The solution for the no transaction costs problem can now be stated as follows.

Theorem 3.1 *Suppose $\Lambda_0 < 0$ and let α_k denote the unique positive solution to (3.1). If (3.2) is satisfied, then the value function for the no transaction costs problem is given by*

$$(3.5) \quad \bar{V}(x, p) = \frac{1}{1-\gamma} \alpha_v p^{-(1-\beta)(1-\gamma)} x^{1-\gamma},$$

and the optimal controls are given in feedback form as

$$(3.6) \quad \bar{C}_t = \alpha_c \bar{X}_t,$$

$$(3.7) \quad \bar{K}_t = \alpha_k \bar{X}_t / P_t,$$

$$(3.8) \quad \bar{\Theta}_t = \alpha_\theta \bar{X}_t,$$

where $\bar{X}_t$ is the wealth process generated by these controls.

The optimal strategy is to use a constant fraction α_c of wealth for consumption of the perishable good, a constant fraction α_k of wealth for consumption of the durable good (at a unit price of P_t), and to keep a constant fraction of wealth invested in each of the financial assets. The optimal controls for the no transaction costs case are hence a natural extension of the optimal controls in the standard Merton setting with power utility from a single perishable consumption good and lognormal asset prices, cf. Merton (1969). Due to substitution effects between the two consumption goods, the fraction of wealth devoted to perishable consumption is generally different from that in Merton's single good setting. The condition (3.2) can be rewritten as

$$(3.9) \quad \rho > (1 - \gamma) \left\{ \frac{1}{2} \gamma \sigma_{P2}^2 \alpha_k^2 + r - (1 - \beta) \mu_P + \frac{1}{2} (1 - \beta) [1 + (1 - \beta)(1 - \gamma)] \|\sigma_P\|^2 \right. \\ \left. + \frac{1}{2\gamma} (\mu - r\mathbf{1} - (1 - \beta)(1 - \gamma) \sigma \sigma_{P1})^\top (\sigma \sigma^\top)^{-1} (\mu - r\mathbf{1} - (1 - \beta)(1 - \gamma) \sigma \sigma_{P1}) \right\},$$

which reduces to the condition on ρ in Merton's problem where $\beta = 1$, and μ_P , σ_{P1} , σ_{P2} , and α_k are all zero.

We see from (3.4) and (3.8) that the vector of fractions of wealth invested in the risky financial assets can be interpreted as a combination of two portfolios, namely the mean-variance tangency portfolio corresponding to $(\sigma \sigma^\top)^{-1} (\mu - r\mathbf{1})$ and a correction or hedge portfolio corresponding to $(\sigma^\top)^{-1} \sigma_{P1}$, i.e. the portfolio having the highest possible absolute correlation between changes in portfolio value and changes in the durable price. The set of risky financial assets therefore exhibits two-fund separation in the sense that any agent is equally well off with access to trade in the two portfolios described above as with access to the complete set of risky financial assets.⁴ In addition to these two funds, the agent will also trade in the riskless financial asset and the two consumption goods.

The correction portfolio appears for two reasons. Firstly, the durable good is a risky investment asset just as the risky financial assets and will, hence, enter the mean-variance portfolio of all risky assets. In (3.4) we have separated out the mean-variance portfolio of the risky financial assets, which is, therefore, to be adjusted for the presence of the remaining risky investment asset, i.e. the durable. Secondly, since the agent extracts utility from the holdings of the durable good, she will seek to hedge changes in the price of the durable by adjusting her financial investment strategy.⁵

To isolate the two components of the correction portfolio term, consider the case where $\beta = 1$, corresponding to the situation where the agent has no direct utility of durable holdings, but still use the durable as an investment asset. Letting $\Psi_t = K_t P_t$ denote the value of the durable holdings, we can rewrite the wealth dynamics (2.5) as

$$\underline{dX_t^{\Theta, \Psi, C} = \left[rX_t^{\Theta, \Psi, C} + (\Theta_t, \Psi_t)^\top \begin{pmatrix} \mu - r\mathbf{1} \\ \mu_P - \delta - r \end{pmatrix} - C_t \right] dt + (\Theta_t, \Psi_t)^\top \begin{pmatrix} \sigma & 0 \\ \sigma_{P1}^\top & \sigma_{P2} \end{pmatrix} dw_t}$$

⁴This two-fund separation result is valid for more general specifications of the instantaneous utility function than the one we assumed in (2.8), but the precise weighting of the two funds in (3.4) is due to that assumption.

⁵If the durable asset only served as an investment asset, the agent would not hedge price changes, since the investment opportunity set is non-stochastic, cf. Merton (1973).

and apply the well-known solution to Merton's problem. Assuming $\mu_P - \delta - r \geq (\mu - r\mathbf{1})^\top (\sigma^\top)^{-1} \sigma_{P1}$, the optimal control policies are of the same form as in Theorem 3.1 but with different multipliers α_c , α_k , and α_θ . In particular,⁶

$$\begin{aligned}\alpha_k &= \frac{1}{\gamma \sigma_{P2}^2} (\mu_P - \delta - r - (\mu - r\mathbf{1})^\top (\sigma^\top)^{-1} \sigma_{P1}), \\ \alpha_\theta &= \frac{1}{\gamma} (\sigma \sigma^\top)^{-1} (\mu - r\mathbf{1}) - (\sigma^\top)^{-1} \sigma_{P1} \alpha_k.\end{aligned}$$

Comparing with (3.4), we see that the presence of the durable good in the instantaneous utility function affects the optimal investment in the financial assets both through the extra term $-(\sigma^\top)^{-1} \sigma_{P1} (1 - \beta) (1 - \gamma)/\gamma$, which reflects the risk aversion and the correlation between the durable price and financial asset prices, and through a change in the optimal investment in the durable asset, i.e. the multiplier α_k .

4 Analytical results for the transaction costs problem

After having solved in closed form the consumption and investment problem for the no transaction costs case, we will next consider the problem with transaction costs on durable trading, i.e. $\lambda > 0$. The presence of transaction costs changes the nature of the problem and it seems impossible to find an explicit solution. Nevertheless, we will in this section derive some important properties of the value function and the optimal strategies by analytic arguments before we turn to the numerical solution of the problem. With the assumed transaction costs on the durable good, the optimal durable trading strategy will be to trade the durable at most countably many times at strictly separated points in time, in contrast to the optimal trading strategy if the transaction costs are proportional to the *change* in the durable holdings, cf. Cuoco and Liu (2000) and the discussion the introduction.⁷ The main result of this section is a complete and simple characterization of the states in which the durable should be traded.

Throughout the section we assume that (3.2) holds and that the initial endowments are such that $(X_{0-}, K_{0-}, P_0) = (x, k, p) \in \bar{\mathcal{S}}$. We will also make the following assumption:

Assumption 4.1 *For all $(x, k, p) \in \bar{\mathcal{S}}$, an optimal policy exists and the value function is finite and satisfies the dynamic programming principle so that for all $\{\mathcal{F}_t\}$ -stopping times τ*

$$(4.1) \quad V(x, k, p) = \sup_{(\Theta, K, C) \in \mathcal{A}(x, k, p)} \mathbb{E} \left[\int_0^\tau e^{-\rho t} U(C_t, K_t) dt + e^{-\rho \tau} V(X_\tau, K_\tau, P_\tau) \right].$$

In particular, let τ correspond to the first time the durable good is traded, so that $K_t = ke^{-\delta t}$ for $t < \tau$, and at time τ there is a decrease in total wealth given by (2.6). The controls (Θ, C) must be chosen such that the solvency condition (2.7) for $t < \tau$ is satisfied with $K_t = e^{-\delta t} k$. We let $\mathcal{A}'(x, k, p)$ denote the set of such controls. Furthermore, the new level of durable holdings, K_τ , must be chosen such that the

⁶We can obtain the same expressions by multiplying (3.1) through by $1 - \beta$ and letting $\beta \rightarrow 1$ in the resulting expression and in (3.4).

⁷In mathematical terms, our optimization problem with transaction costs is a combined stochastic control and impulse control problem, cf. Chancelier, Øksendal, and Sulem (2000) and Korn (1997, Ch. 5).

solvency condition continues to hold, i.e. $K_\tau \leq X_{\tau-}/(\lambda P_\tau) - ke^{-\delta\tau}$. It follows from (4.1) that

$$(4.2) \quad V(x, k, p) = \sup_{\substack{(\Theta, C) \in \mathcal{A}'(x, k, p), \\ K_\tau \leq \frac{X_\tau}{\lambda P_\tau} - ke^{-\delta\tau}}} \mathbb{E} \left[\int_0^\tau e^{-\rho t} U(C_t, ke^{-\delta t}) dt + e^{-\rho\tau} V(X_{\tau-} - \lambda ke^{-\delta\tau} P_\tau, K_\tau, P_\tau) \right].$$

We shall apply this equation in the reduction of the dimensionality of the problem below.

4.1 Properties of the value function

The following theorem gives some important basic properties of the value function. The proof is given in Appendix B.

Theorem 4.1 *The value function $V(x, k, p)$ satisfies:*

(a) *For all $(x, k, p) \in \bar{\mathcal{S}}$, we have that*

$$\frac{1}{1-\gamma} \underline{\alpha} p^{-(1-\beta)(1-\gamma)} (x - \lambda kp)^{1-\gamma} \leq V(x, k, p) \leq \frac{1}{1-\gamma} \alpha_v p^{-(1-\beta)(1-\gamma)} x^{1-\gamma},$$

where

$$\underline{\alpha} = \frac{\beta^{\beta(1-\gamma)} (1-\beta)^{(1-\beta)(1-\gamma)} r^{\beta(1-\gamma)}}{\rho + \delta(1-\beta)(1-\gamma)}$$

and where α_v is given by (3.3).

(b) *For each $(k, p) \in \mathbb{R}_{++}^2$, $V(x, k, p)$ is strictly increasing and concave in x on $\{x : x \geq \lambda kp\}$.*

(c) *For each $(k, p) \in \mathbb{R}_{++}^2$, $V(x, k, p)$ is continuous in x on $\{x : x > \lambda kp\}$.*

(d) *$V(x, k, p)$ is homogeneous of degree $1-\gamma$ in (x, k) and of degree $\beta(1-\gamma)$ in (x, p) , i.e. for all $(x, k, p) \in \bar{\mathcal{S}}$ and all $\kappa > 0$*

$$(4.3) \quad V(\kappa x, \kappa k, p) = \kappa^{1-\gamma} V(x, k, p),$$

$$(4.4) \quad V(\kappa x, k, \kappa p) = \kappa^{\beta(1-\gamma)} V(x, k, p).$$

As discussed immediately after Theorem 4.2, $V(x, k, p)$ is in fact continuous on the larger set $\{(x, k, p) \in \mathbb{R}_{++}^3 : x \geq \lambda kp\}$.

Because of the homogeneity properties (4.3) and (4.4) of the value function, we can reduce the problem of finding V on the set $\mathbb{R}_{++}^3 \cap \mathcal{S}$ from one with three state variables (x, k, p) to one with a single state variable, namely the ratio $z = x/(kp)$ of total wealth to durable wealth, since

$$(4.5) \quad V(x, k, p) = k^{1-\gamma} p^{\beta(1-\gamma)} v(x/(kp)), \quad (x, k, p) \in \mathbb{R}_{++}^3 \cap \mathcal{S},$$

where v is defined as

$$(4.6) \quad v(z) = V(z, 1, 1), \quad z \in \bar{\mathcal{S}}_1,$$

and $\mathcal{S}_1 = (\lambda, \infty)$. In particular, $v(\lambda) = V(\lambda, 1, 1) = 0$, since the only feasible strategy with the initial conditions $(x, k, p) = (\lambda, 1, 1)$ is to sell the unit of durable, which leaves the agent with zero wealth and

zero consumption in all future. From Theorem 4.1, we know that v is strictly increasing and concave on $[\lambda, \infty)$, continuous on (λ, ∞) , and that

$$(4.7) \quad \frac{1}{1-\gamma} \underline{\alpha}(z-\lambda)^{1-\gamma} \leq v(z) \leq \frac{1}{1-\gamma} \alpha_v z^{1-\gamma}.$$

Define the process $Z_t = X_t/(K_t P_t)$ with initial value $Z_0 = z = x/(kp)$ and introduce the scaled controls $\hat{C}_t = C_t/(P_t K_t)$ and $\hat{\Theta}_t = \Theta_t/(P_t K_t)$. Note that $\hat{C}_t/Z_t = C_t/X_t$ and $\hat{\Theta}_t/Z_t = \Theta_t/X_t$. Substituting (4.5) into (4.2), we get

$$(4.8) \quad p^{\beta(1-\gamma)} v(z) = \sup_{(\hat{\Theta}, \hat{C}) \in \hat{\mathcal{A}}(z), \tau \in \mathcal{T}} \mathbb{E} \left[\frac{1}{1-\gamma} \int_0^\tau e^{-\bar{\rho}t} \hat{C}_t^{\beta(1-\gamma)} P_t^{\beta(1-\gamma)} dt + e^{-\bar{\rho}\tau} P_\tau^{\beta(1-\gamma)} (Z_{\tau-} - \lambda)^{1-\gamma} \frac{M}{1-\gamma} \right],$$

where $\bar{\rho} = \rho + \delta(1-\gamma)$,

$$M = \sup_{K_\tau \leq k e^{-\delta\tau} (Z_{\tau-} - \lambda)/\lambda} (1-\gamma) \left(\frac{k e^{-\delta\tau} (Z_{\tau-} - \lambda)}{K_\tau} \right)^{\gamma-1} v \left(\frac{k e^{-\delta\tau} (Z_{\tau-} - \lambda)}{K_\tau} \right) = (1-\gamma) \sup_{z \geq \lambda} z^{\gamma-1} v(z),$$

and $\hat{\mathcal{A}}(z)$ is the set of $(\hat{\Theta}, \hat{C}) \in \mathcal{L}^2 \times \mathcal{C}$ with $Z_t = Z_t^{\hat{\Theta}, \hat{C}} \geq \lambda$ for all $t \geq 0$. The problem can now be seen as a combined stochastic control and optimal stopping problem with the stopping value $v(z) = f(z)$, where $f(z) = (z-\lambda)^{1-\gamma} M/(1-\gamma)$ is the stopping reward function.⁸ It follows from (4.7) that $\underline{\alpha} \leq M \leq \alpha_v$. Also note that

$$(4.9) \quad z^* = \arg \max_{z \geq \lambda} z^{\gamma-1} v(z)$$

is the value of the transformed state variable after the optimal change in the durable holdings at time τ , since

$$Z_\tau = \frac{X_\tau}{K_\tau P_\tau} = \frac{X_{\tau-} - \lambda K_{\tau-} P_\tau}{K_\tau P_\tau} = \frac{K_{\tau-}}{K_\tau} (Z_{\tau-} - \lambda) = z^*.$$

The new level of durable holdings can be expressed in terms of z^* as $K_\tau = K_{\tau-} (Z_{\tau-} - \lambda)/z^*$.

The HJB equation associated with the reduced problem can be written as⁹

$$(4.10) \quad 0 = \max \left\{ H(z, v, v', v''), (z-\lambda)^{1-\gamma} \frac{M}{1-\gamma} - v(z) \right\}, \quad \forall z \in \mathcal{S}_1,$$

where

$$(4.11) \quad H(z, v, v', v'') = \sup_{\hat{\theta} \in \mathbb{R}^n, \hat{c} \in \mathbb{R}_{++}} \left\{ \mathcal{L}^{\hat{\theta}, \hat{c}} v(z) + \frac{1}{1-\gamma} \hat{c}^{\beta(1-\gamma)} - \bar{\rho} v(z) \right\}$$

⁸Note, however, that M itself depends on the value function $v(z)$, and, therefore, the problem is not a standard optimal stopping problem. In this case the function $f(z)$ is elsewhere referred to as the intervention value. See footnote 7 and the references given there.

⁹The HJB equation can be derived by substituting (4.5) into the HJB equation for the original problem (derived under the assumption that $V(x, k, p)$ is twice continuously differentiable). Alternatively, we can use (4.8) and derive the dynamics of the process $P_t^{\beta(1-\gamma)} v(Z_t)$ through an application of Itô's Lemma.

with the operator $\mathcal{L}$ given by

(4.12)

$$\begin{aligned} \mathcal{L}^{\hat{\theta}, \hat{c}} v(z) = & v(z) \beta(1 - \gamma) \left[\mu_P - \frac{1}{2} [1 - \beta(1 - \gamma)] \|\sigma_P\|^2 \right] \\ & + v'(z) \left[(z - 1)(r - \mu_P + \delta + [1 - \beta(1 - \gamma)] \|\sigma_P\|^2) - \hat{c} + \hat{\theta}^\top (\mu - r\mathbf{1} - [1 - \beta(1 - \gamma)] \sigma \sigma_{P1}) \right] \\ & + \frac{1}{2} v''(z) \left[\hat{\theta}^\top \sigma \sigma^\top \hat{\theta} - 2(z - 1) \hat{\theta}^\top \sigma \sigma_{P1} + (z - 1)^2 \|\sigma_P\|^2 \right]. \end{aligned}$$

The reduced value function $v(z)$ is not necessarily smooth and, therefore, it cannot be expected to satisfy the HJB equation (4.10) in the classical sense. We show below that v is a solution in a generalized sense, namely a so-called *viscosity solution*. While the precise definition of viscosity solutions of (4.10) is given in Appendix B, we note here that a viscosity solution of (4.10) is continuous on $\mathcal{S}_1$, but not necessarily differentiable.¹⁰ For other applications of viscosity theory to optimal consumption and investment problems, see, e.g., Fleming and Zariphopoulou (1991), Davis, Panas, and Zariphopoulou (1993), Duffie et al. (1997), Hindy, Huang, and Zhu (1997), and Constantinides and Zariphopoulou (1999). The next theorem shows that the reduced value function v is a viscosity solution of (4.10) and that if the parameter constraint

$$(4.13) \quad \rho > (1 - \gamma) \left(\beta \mu_P - \frac{1}{2} \beta [1 - \beta(1 - \gamma)] \|\sigma_P\|^2 - \delta \right)$$

holds, then v is the only viscosity solution of (4.10) satisfying some desired properties. The proof is relegated to Appendix B.

Theorem 4.2 *The function v defined by (4.6) has the following properties:*

- (a) *v is a viscosity solution of the HJB equation (4.10).*
- (b) *v is continuous on $[\lambda, \infty)$.*
- (c) *if (4.13) holds, then v is the only strictly increasing and concave viscosity solution of (4.10) which is continuous on $[\lambda, \infty)$ and satisfies $v(\lambda) = 0$.*

Since $V(x, k, p) = k^{1-\gamma} p^{\beta(1-\gamma)} v(x/(kp))$ for all $(x, k, p) \in \mathbb{R}_{++}^3$ with $x \geq \lambda kp$, (b) implies that $V(x, k, p)$ is continuous on the set $\{(x, k, p) \in \mathbb{R}_{++}^3 : x \geq \lambda kp\}$.

4.2 Properties of the optimal consumption and investment strategies

The solvency region $\mathcal{S}$ can be divided into a region where it is optimal not to trade the durable, a region where it is optimal to reduce durable holdings, and a region where it is optimal to increase durable holdings. In terms of the state variable z of the reduced problem, the no-trade region (continuation region) is the set

$$\hat{\mathcal{N}} = \{z > \lambda : v(z) > f(z)\},$$

¹⁰The viscosity solution concept can even be further generalized to discontinuous functions, cf. Chancelier, Øksendal, and Sulem (2000).

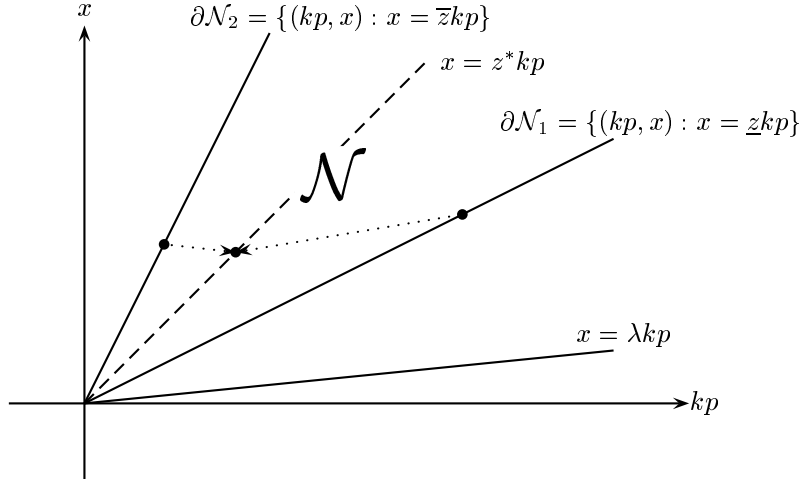


Figure 1: Illustration of the partitioning of the state space.

where $f(z) = (z - \lambda)^{1-\gamma}M/(1 - \gamma)$ is the stopping reward function. The next theorem shows that the no-trade region $\hat{\mathcal{N}}$ is an open interval $(\underline{z}, \bar{z})$ and that the value of z immediately after a durable trade, z^* , is in the open interval $(\underline{z}, \bar{z})$. The proof, which applies that v is a viscosity solution of the HJB equation (4.10), is given in Appendix B.

Theorem 4.3 *If (4.13) holds and $H(z, f, f', f'') > 0$ for some $z \in (\lambda, \infty)$, then numbers $\underline{z}$ and $\bar{z}$ exist satisfying $\lambda < \underline{z} < \bar{z}$ such that $\hat{\mathcal{N}} = (\underline{z}, \bar{z})$. Also, z^* defined by (4.9) satisfies $z^* \in (\underline{z}, \bar{z})$.*

The no-trade interval $\hat{\mathcal{N}}$ corresponds to the cone

$$\mathcal{N} = \left\{ (kp, x) : \underline{z} < \frac{x}{kp} < \bar{z} \right\}$$

in the (kp, x) -plane as illustrated in Figure 1. At and below the lower boundary $\partial\mathcal{N}_1$ of $\mathcal{N}$, the value of the durable holdings is so high relative to the total wealth that it is optimal to decrease the durable holdings. At and above the upper boundary $\partial\mathcal{N}_2$ of $\mathcal{N}$, the value of the durable holdings is so low relative to the total wealth that it is optimal to increase the durable holdings. In both cases, the optimal position immediately after the durable trade is a point on the dashed line where $x = z^*kp$. As depicted in Figure 1, this point can be found as the crossing point between the dashed line and a straight line through the point corresponding to the position just before the durable trade. For a point on the lower boundary $\partial\mathcal{N}_1$ the straight line has a slope of $\lambda z^*/(z^* - \underline{z} + \lambda) > 0$ and for a point on the upper boundary $\partial\mathcal{N}_2$ the slope is $\lambda z^*/(z^* - \bar{z} + \lambda) < 0$. Note that from (3.7) the optimal relation between total wealth and durable wealth without transaction costs is given by $X_t/(K_t P_t) = 1/\alpha_k$, which will generally be different from z^* and may fall outside the no-trade interval $(\underline{z}, \bar{z})$ for some parameter combinations (see the second line of Part C of Table 1 in Section 5).

We can now conclude that if the initial values are such that $(kp, x) \notin \mathcal{N}$, the optimal durable trading strategy will involve an initial trade to the line $x = z^*kp$. Thereafter, the durable is only traded when the

evolution of uncertain prices is such that $(K_t - P_t, X_t)$ hits one of the boundaries of $\mathcal{N}$ in which case the durable good is traded back to the relation $X_t = z^* K_t P_t$. The strategy therefore consists of infrequent, non-infinitesimal durable transactions contrary to the findings in models with proportional transaction costs to financial asset transactions, cf., e.g., Davis and Norman (1990), and in the model of divisible durable goods studied by Cuoco and Liu (2000). In those models, the optimal transactions are of a local time nature and involves frequent, infinitesimal trades.

We close this section with a brief discussion of the optimal investment strategies in the financial assets. Assuming v is twice differentiable, the first order conditions for the scaled control values in (4.11) are

$$(4.14) \quad \hat{\theta} = -\frac{v'(z)}{v''(z)} (\sigma \sigma^\top)^{-1} (\mu - r\mathbf{1} - [1 - \beta(1 - \gamma)] \sigma \sigma_{P,1}) + (\sigma^\top)^{-1} \sigma_{P,1} (z - 1),$$

$$(4.15) \quad \hat{c} = (\beta^{-1} v'(z))^{\frac{1}{\beta(1-\gamma)-1}}.$$

In particular, we can write the vector of fractions of wealth optimally invested in the risky financial assets as

$$\frac{\theta}{x} = \frac{\hat{\theta}}{z} = -\frac{v'(z)}{z v''(z)} (\sigma \sigma^\top)^{-1} (\mu - r\mathbf{1}) - \left(-\frac{v'(z)}{z v''(z)} (1 - \beta[1 - \gamma]) + \frac{1}{z} - 1 \right) (\sigma^\top)^{-1} \sigma_{P,1}.$$

Comparing this expression with the constant vector α_θ in (3.4), which by Theorem 3.1 gives the optimal fractions invested in the risky financial assets without transaction costs, we see that the optimal investment strategy again combines the mean-variance portfolio, represented by $(\sigma \sigma^\top)^{-1} (\mu - r\mathbf{1})$, and the portfolio hedging changes in the price of the durable good, represented by $(\sigma^\top)^{-1} \sigma_{P,1}$. The weights of the funds are very similar to the no costs case. Firstly, $1/\gamma$ in (3.4) is replaced by $-v'(z)/(z v''(z))$, but (by Theorem 3.1) $1/\gamma$ is the relative risk tolerance of the agent for the no costs case and $-v'(z)/(z v''(z)) = -\frac{\partial V}{\partial x}(x, k, p) / (x \frac{\partial^2 V}{\partial x^2}(x, k, p))$ is the relative risk tolerance with transaction costs, which varies with z as we will see from the numerical results in Section 5. Secondly, the constant α_k in (3.4), denoting the optimal fraction of wealth held in the durable without transaction costs, is replaced by $1/z = kp/x$, which is the current fraction of wealth held in the durable good.

5 Numerical results for the transaction costs problem

To further investigate optimal behavior in our setting we resort to numerical examples. We solve the HJB equation (4.10) numerically with the iterative scheme outlined in Appendix C. While a complete proof of convergence of this method is beyond the scope of this paper, we note that the central requirement for convergence is the existence of a unique viscosity solution to the HJB equation, which is provided by Theorem 4.2. Also, our approach is closely related to the method proposed by Chancelier, Øksendal, and Sulem (2000) for a similar problem for which convergence to the viscosity solution of the HJB equation is guaranteed. In our numerical examples we fix the number of risky financial securities to two, which is identical to the number of separating risky funds. We assume the following parameter values throughout

the numerical part:

$$r = 0.05, \quad \mu = \begin{pmatrix} 0.1 \\ 0.15 \end{pmatrix}, \quad \rho = 0.2, \quad \delta = 0.04.$$

Furthermore, the individual stock volatilities are fixed to 0.25 and 0.3, respectively.

5.1 Optimal behavior as a function of the state

In this subsection, we consider how consumption and portfolio choice change as the ratio z of total wealth to wealth in durable changes. For this investigation we assume that the two stocks are uncorrelated and that each stock has a positive correlation with the durable of 0.5. Furthermore, we assume that

$$\mu_P = 0.07, \quad \|\sigma_P\| = 0.12, \quad \beta = 0.5, \quad \gamma = 0.5, \quad \lambda = 0.05.$$

Figure 2 shows the difference between the value function, $v(z)$, and the value if a transaction of the durable is performed, i.e. the stopping reward function $M(z - \lambda)^{1-\gamma}/(1 - \gamma)$. The shape of the curve is very similar to that of the corresponding difference in Grossman and Laroque (1990). Close to the boundaries of the no-trade region the value function is less concave than $M(z - \lambda)^{1-\gamma}/(1 - \gamma)$, while it is more concave between the two boundaries. This implies that the agent is more risk averse in the interior of the no-trade region than close to the boundaries. This is illustrated in Figure 3.

In Figure 3, we plot the relative risk aversion associated with the indirect utility of total wealth x , i.e. $-x \frac{\partial^2 V}{\partial x^2}(x, k, p) / \frac{\partial V}{\partial x}(x, k, p) = -zv''(z)/v'(z)$. The figure confirms that risk aversion changes over the no-trade region. Near $\underline{z}$ or $\bar{z}$ the agent is less risk averse than if transaction costs were absent, whereas the agent is more risk averse in the middle of the region. When the state is close to z^* , the loss in utility associated with transacting the durable is large, cf. Figure 2, and the agent attempts to avoid new durable purchases and, therefore, behaves in a more risk averse manner. Closer to the boundaries of the no-trade region, the monetary loss associated with the potential transaction costs is compensated for by a change to the optimal ratio of total wealth to durable wealth, and the agent is therefore much less risk averse.

This change in risk aversion has a considerable impact on consumption and portfolio choice. In Figure 4, we plot the fraction of wealth optimally invested in the riskfree security and the two separating risky funds as a function of z . Fund 1 is the tangency portfolio $(\sigma\sigma^\top)^{-1}(\mu - r\mathbf{1})$, the portfolio with highest possible Sharpe ratio, and Fund 2 is the portfolio $(\sigma^\top)^{-1}\sigma_P$ that optimally hedges changes in the price of the durable. The optimal portfolio choices are quite different from the no transaction costs case of Section 3, where the fraction of wealth invested in each fund is constant. Not surprisingly, the lower risk aversion close to the boundaries leads to higher fractions of the wealth placed in the tangency portfolio when the state variable is close to the boundary of the no-trade region than when the state variable is in the center of the no-trade region. The stocks are positively correlated with the durable price, so durable price changes are hedged if the fraction of wealth in the hedge portfolio is negative, hence the fraction of wealth placed in Fund 2 is negative. Close to the lower bound $\underline{z}$, durable wealth is very high compared

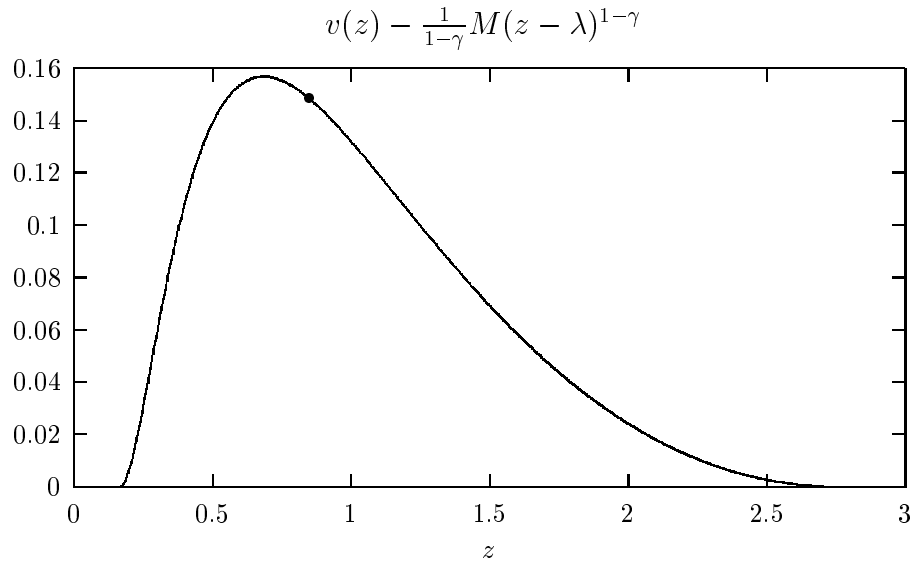


Figure 2: The difference between the value function and the value of changing durable level is plotted against $z \in (\underline{z}, \bar{z}) = (0.17, 2.75)$. The $\bullet$ represents the location just after durable purchase, where $z^* = 0.85$. Without transaction costs, the optimal durable holdings are such that $x/(kp) = 0.51$.

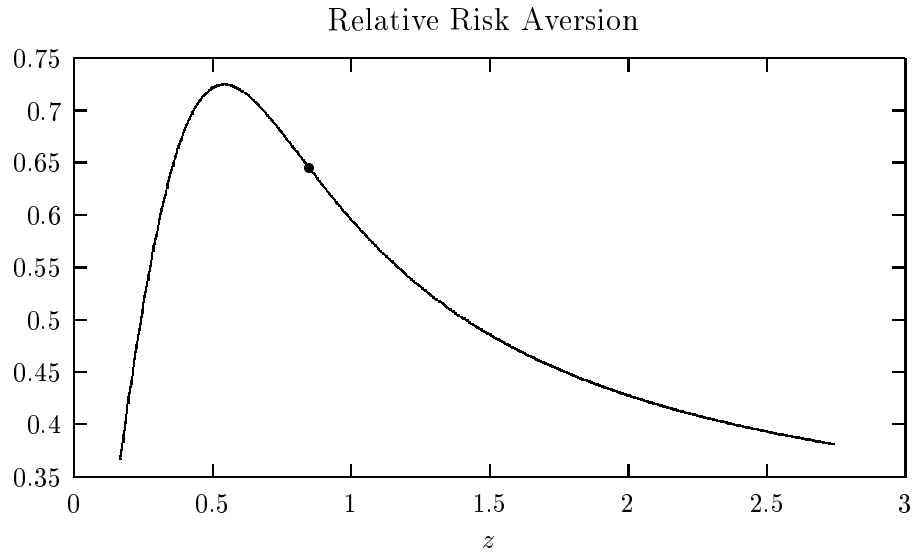


Figure 3: The relative risk aversion is plotted against $z \in (\underline{z}, \bar{z})$. The $\bullet$ indicates the point just after durable purchase, where relative risk aversion is 0.65. The relative risk aversion without transaction costs is 0.5.

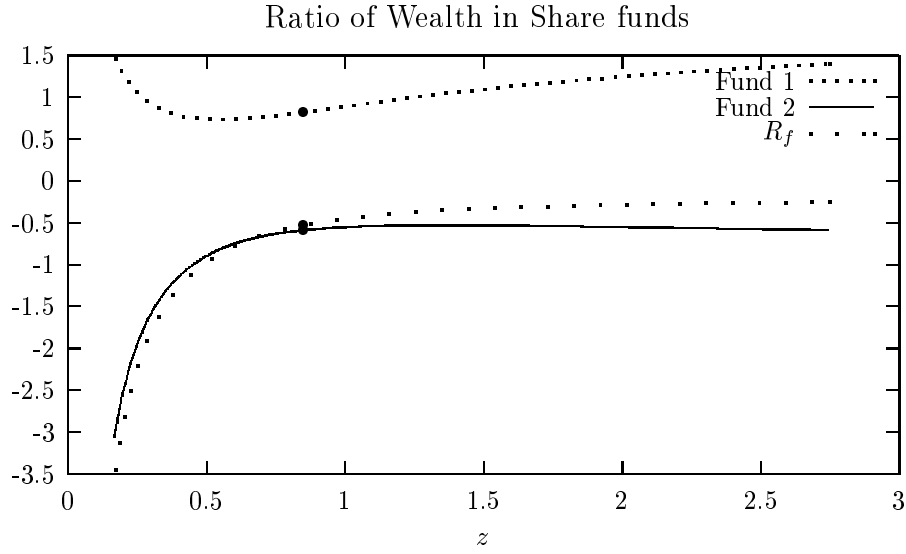


Figure 4: Fractions of total wealth placed in the funds. Fund 1 is the tangency portfolio and Fund 2 is the portfolio that hedges changes in the price on the durable good. The • represents the optimal position just after transacting the durable. Finally, the dotted curve labeled R_f represents the ratio of wealth in the riskfree security.

to total wealth, and the wealth placed in the portfolio hedging durable price changes is therefore also high, in absolute terms, compared to total wealth. For completeness, the fraction of wealth placed in the riskless security is plotted. The agent will always borrow for the preferences and risk premia in this numerical example. Moreover, with the parameter values assumed, the target wealth ratio z^* is smaller than one corresponding to a negative financial wealth, i.e. the agent finances the durable holdings with a (net) short position in the financial market.

In Figure 5, the optimal propensity to consume, $c/x = \hat{c}/z$, is plotted against z . Without transaction costs, a constant fraction of wealth is used for consumption of the perishable consumption good, and a constant fraction of wealth is used for consumption of the durable good. These results are implied by the first order conditions from the consumption-investment problem, where the marginal utility of perishable consumption equals both the marginal utility of wealth and the marginal utility of durable consumption. With transaction costs, it is infinitely expensive, and hence inoptimal, to keep these ratios aligned. Assuming for a moment that v is twice continuously differentiable, it can be shown that the propensity to consume is increasing in z if and only if $-zv''(z)/v'(z) > -c\frac{\partial^2 U}{\partial c^2}(c, k)/\frac{\partial U}{\partial c}(c, k) = 1 - \beta(1 - \gamma)$, i.e. whenever the relative risk aversion with respect to wealth changes is higher than the “relative risk aversion” with respect to changes in the perishable consumption flow. For $\gamma = 0.5$ the relative risk aversion with respect to wealth changes is everywhere below $1 - \beta(1 - \gamma) = 0.75$, cf. Figure 3, so the propensity to consume is in this case decreasing in z , consistent with the solid line in Figure 5. To illustrate a case where the propensity to consume is non-decreasing, we have included the dotted line, where $\gamma = 0.7$. The bullets indicate consumption rates just after a durable transaction.

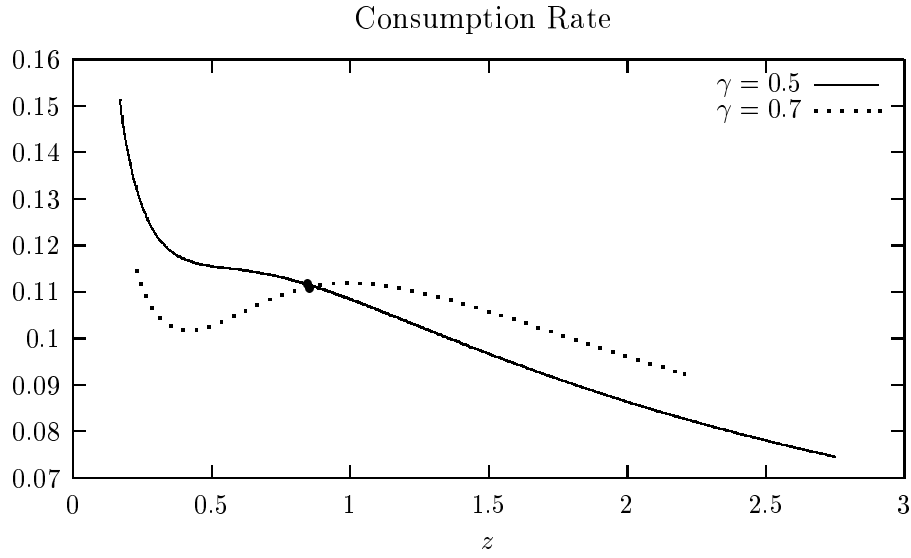


Figure 5: The ratio of perishable consumption to total wealth plotted against z . The solid line is for the benchmark case, $\gamma = 0.5$. Furthermore, the dotted line represents the case where γ is changed to 0.7. The $\bullet$'s represent choices immediately after transacting the durable.

Next, we will consider how the expected time to next durable transaction varies with the state. If we let $u(z)$ denote the expected time it takes for Z_t to reach either $\underline{z}$ or $\bar{z}$ given that $Z_0 = z$, it follows from Karlin and Taylor (1981, Sec. 15.3) that u satisfies the ordinary differential equation (ODE)

$$\frac{1}{2}\sigma_Z(z)^2 u''(z) + \mu_Z(z)u'(z) = -1 \quad \text{for } \underline{z} < z < \bar{z}, \quad u(\underline{z}) = u(\bar{z}) = 0,$$

where μ_Z and σ_Z are the drift and diffusion terms, respectively, of the optimally controlled state process $Z_t = X_t/(K_t P_t)$. This ODE is solved numerically by a standard method. Figure 6 shows that the expected waiting time function has a similar shape as the utility gain from not trading the durable depicted in Figure 2. Even with the moderate transaction costs level assumed in this example ($\lambda = 0.05$), the expected time between two durable transactions is 3.58 years. Nevertheless, the expected waiting times we find are generally significantly lower than those found by Grossman and Laroque (1990), because to avoid transaction costs the agent can in our model substitute perishable consumption for durable consumption. Consequently, the optimal fraction of wealth placed in the durable consumption good is lower than if there were no perishable good. This decrease in the stock of the durable good effectively decreases the impact of the transaction costs, leading to more frequent trade in the durable than in the single-good model of Grossman and Laroque.

5.2 Parameter sensitivity

In this subsection, we study how optimal behavior changes as the parameters of the market change. We emphasize the effects that cannot be studied in the simpler model of Grossman and Laroque (1990).

As discussed above, the agent can substitute perishable consumption for durable consumption to

Expected Waiting Time

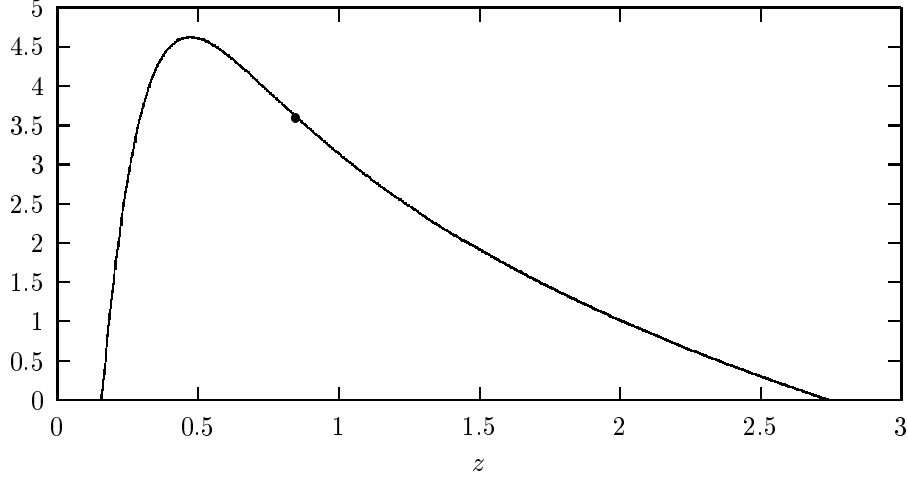


Figure 6: The expected time (in years) to the next durable transaction as a function of the current state. The • represents the expected waiting time immediately after a durable transaction, i.e. at z^* .

reduce the transaction costs implied by a durable trade. Shifting the importance of durable consumption relative to perishable consumption (shifting β) in the direct utility function, as we do in Part A of Table 1, demonstrates the importance of the alternative good. For low values of β , the durable good has a higher weight in the utility function, making the agent more vulnerable to changes in durable level. Moreover, increasing β increases the perishable consumption rate and decreases the durable wealth to total wealth ratio, thereby decreasing the costs associated with transacting the durable, resulting in more frequent trading of the durable and increasing the probability of buying more of the durable good at next trade.¹¹

Turning to the importance of the durable price process, we first note that in the benchmark case the durable consumption good is not attractive as an investment object. Both risky financial securities have a higher expected return to volatility ratio; there are transaction costs associated with selling the durable; the durable depreciates physically; and the durable is positively correlated with the financial securities, so although the durable provides further diversification, hedging unwanted changes in the durable level is associated with a decrease in expected return on financial securities.

Figure 7 and Part B of Table 1 illustrate how changes in the correlation between the financial asset prices and the durable price affect the optimal behavior. The optimal level of the state variable, z^* , is an increasing function of the durable-stock correlation, corresponding to lower durable holdings for higher correlation. Similarly, the lower bound on the continuation region is increasing in the durable-stock correlation, which is also the case for the upper bound up to a certain, positive correlation after which the upper bound decreases with the correlation. To understand these results note that there are

¹¹The probability $q(z)$ that Z_t reaches $\bar{z}$ before $\underline{z}$ given that the current state is $Z_0 = z$ can be found as the solution to the ODE $\frac{1}{2}\sigma_Z(z)^2 q''(z) + \mu_Z(z)q'(z) = 0$, $z \in (\underline{z}, \bar{z})$, with the boundary conditions $q(\underline{z}) = 0$, $q(\bar{z}) = 1$, cf. Karlin and Taylor (1981, Sec. 15.3).

	$\underline{z}$	$\bar{z}$	$\frac{1}{z^*}$	α_k	RRA	c^*	α_c	$E(\tau)$	$P(z_\tau = \bar{z})$
Benchmark	0.17	2.75	1.16	1.96	0.64	0.096	0.12	3.58	0.94
β	Part A: Changes in β								
0.3	0.13	1.99	1.61	2.71	0.71	0.043	0.07	4.13	0.89
0.7	0.26	4.53	0.69	1.17	0.58	0.218	0.16	3.04	0.96
Corr.	Part B: Changes in durable/stock correlation								
-0.5	0.07	1.98	2.41	6.13	0.67	0.035	0.07	4.96	0.44
0	0.13	2.69	1.35	2.49	0.63	0.074	0.10	4.42	0.72
$\mu_P/\ \sigma_P\ $	Part C: Changes in durable risk/return relationship								
0.1/0.05	0.06	1.36	3.38	10.65	0.76	0.031	0.11	6.52	0.87
0.09/0.01	0.05	1.15	4.91	33.03	0.78	0.021	0.12	11.21	0.93
λ	Part D: Changes in transaction costs								
0.002	0.27	1.05	1.73	1.96	0.53	0.068	0.12	0.96	0.75
0.2	0.26	5.58	0.78	1.96	0.74	0.138	0.12	5.00	0.99

Table 1: Columns 1 and 2 are the lower and upper bound on the no-trade region. Column 3 is the optimal durable wealth/total wealth ratio immediately after a durable trade. Column 4 is the corresponding ratio without transaction costs. Columns 5 and 6 are the relative risk aversion and optimal consumption rate just after durable trade, respectively, and Column 7 represents the optimal consumption rate without transaction costs. Columns 8 and 9 are the expected waiting time until the next durable trade and the probability of trading to a higher durable level at next durable trade, estimated just after a durable trade. The first row in the table represents the benchmark case, described previously in this section. The four parts illustrate deviations from the benchmark in alternative directions. In Part A, the relative importance of the durable vs. the perishable good is changed. In Part B, changes in the durable/stock correlation are considered. Part C considers changing the volatility and expected return on the price of the durable good, and, finally in Part D, we change the transaction costs associated with trading the durable.

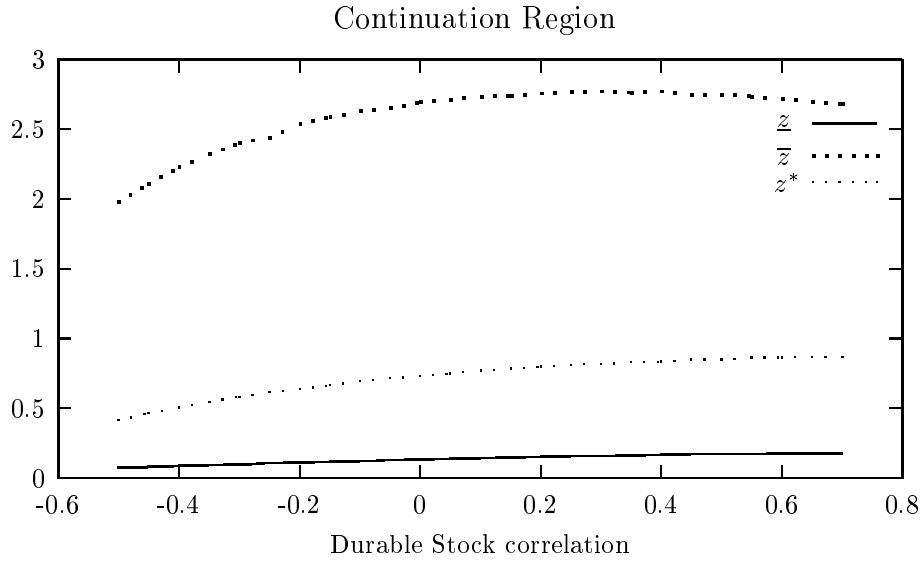


Figure 7: The change in the no-trade region as the correlation between the durable consumption good and each of the risky securities changes.

several consequences of varying the correlation coefficient. Firstly, as the correlation increases, the durable becomes less effective as an instrument to diversify away financial risk, which should lead to lower durable holdings. Secondly, varying the correlation from zero towards either +1 or -1 makes it easier to hedge the price risk of the durable, which, other things equal, lead to lower durable holdings for zero correlation than for either positive or negative correlation. Thirdly, if the correlation is positive (negative), the hedge portfolio consists of negative (positive) positions in the stocks, so hedging the durable is associated with a wealth loss (gain), because the return on the hedge portfolio with the assumed parameters is higher than the return on the durable. All three effects support a decrease in optimal durable holdings (an increase in z^*) as the durable-stock correlation is increased from negative values towards zero. As the correlation is further increased to positive values, the second effect tends to increase durable holdings. In our example it does not dominate the two other effects at the optimal level, but it does so at the upper boundary of the no-trade region which decreases from a certain value of the correlation coefficient. The size of the no-trade region is decreasing in the absolute size of correlation since higher absolute correlation increases the possibility of hedging durable price changes, and it is therefore much easier to keep the state variable close to the optimal level. Decreasing the durable/stock correlations increases the optimal level of durable, thereby increasing the costs associated with trading the durable. It is therefore optimal to trade the durable less frequently so that the expected waiting time to the next durable trade increases. Furthermore, it becomes more likely that the next trade of durable is to a lower level.

In Part C of Table 1, we vary the expected return to volatility ratio of the durable good. With a smaller price volatility the optimal holdings of the durable are substantially higher, the no-trade region is narrower, and the expected period of time between durable trades is significantly longer. Also note that the impact the assumed transaction costs has on the optimal durable wealth to total wealth ratio

depends heavily on the risk/return relationship of the durable good. In the last case in Part C, where the durable good is quite attractive for investment purposes, transaction costs have the dramatic effect of lowering the optimal durable wealth to total wealth ratio by approximately 85% (from 33.03 to 4.91).

Finally, we consider the sensitivity of our results to changes in the transaction costs parameter λ . In Figure 8, the no-trade region is plotted for varying transaction costs together with z^* , the value of the state variable just after trading the durable. Even very small transaction costs lead to a relatively wide no-trade region. Without transaction costs z^* is 0.51, whereas with $\lambda = 0.002$ the no-trade region is $(0.27, 1.05)$. As λ increases, the size of the continuation region increases in accordance with the intuition that higher transaction costs lead the agent to trade the durable more infrequently. Part D of Table 1 shows that with small costs of $\lambda = 0.002$ the expected waiting time to the next durable trade is almost 1 year. Figure 9 provides a close-up of the lower bound of the no-trade region. For low levels of transaction costs, an increase in transaction costs makes the investor more reluctant to trade in the durable, causing $\underline{z}$ to decrease. As the transaction costs increase, the solvency region becomes smaller, and the potential threat of bankruptcy naturally drives $\underline{z}$ up. For very high transaction costs, e.g. $\lambda = 1$, the durable holdings are reduced because an adverse move in the durable price may otherwise lead to bankruptcy.

The optimal wealth ration with transaction costs, z^* , varies from 0.57 for $\lambda = 0.002$ to 1.37 for $\lambda = 0.25$, and it is therefore everywhere higher than z^* without transaction costs, i.e. the optimal durable holdings are lower with transaction costs. The choice of z^* with transaction costs is influenced by several factors in addition to those without transaction costs. To compensate for the physical depreciation of the durable and the expected wealth increase until the agent can trade the durable again, the consumer/investor tends to purchase a higher level of durable than if there were no transaction costs. On the other hand, the transaction costs when the durable is traded are increasing in the durable position, and this tends to decrease the durable holdings of the agent. In our example, the latter effect dominates and the durable purchases are lower than in the no transaction costs case. Furthermore, as can be seen from Part D of Table 1, as the transaction costs increase, durable consumption is shifted to perishable consumption and the agent will act in a more risk averse manner.

6 Concluding remarks

We have studied an optimal consumption and investment problem in a setting merging the classical models of Merton (1969) and Grossman and Laroque (1990) so that the agent extracts utility from consumption of both a perishable and a durable consumption good. For the case where the durable good can be traded without transaction costs, we have derived an explicit Merton-style solution to the problem. With transaction costs, we used the notion of viscosity solutions to show that optimal behavior is determined by the ratio z of total wealth to wealth in the durable and that as long as z is in the interval $(\underline{z}, \bar{z})$, the durable is not transacted. At the boundaries, the durable stock is adjusted such that the new value of the ratio z equals a $z^* \in (\underline{z}, \bar{z})$. Furthermore, we have illustrated by numerical examples several important economic effects that do not appear in the simpler one-good models. Our results demonstrate

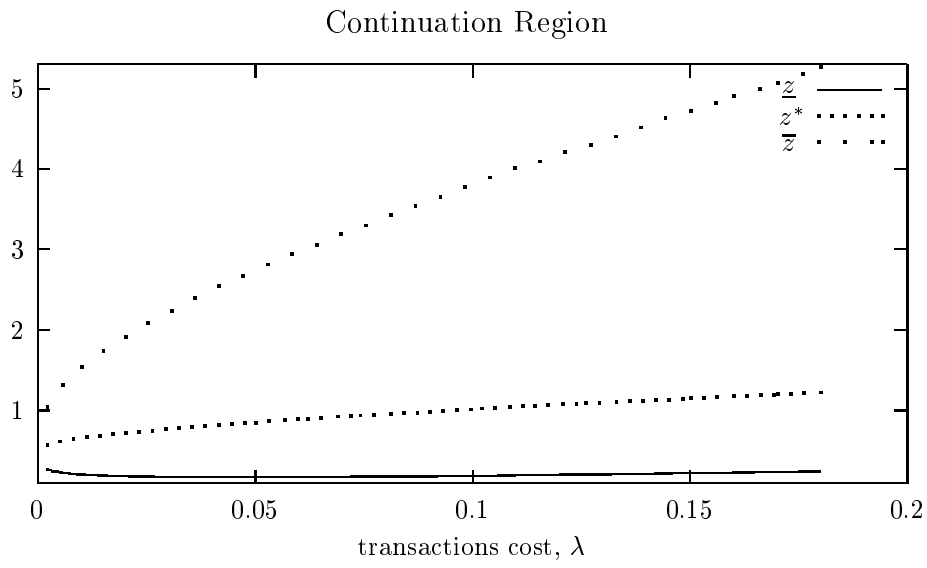


Figure 8: The change in the no-trade region as transaction costs change.

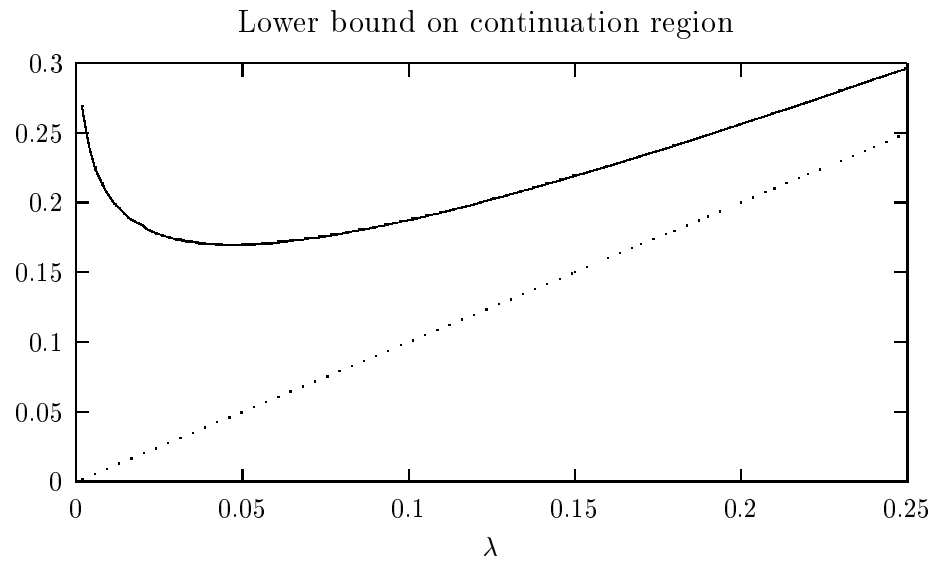


Figure 9: The solid line is a plot of the lower bound for no-trade region against the transaction costs parameter λ . The dotted line is the boundary of the solvency region.

that the optimal perishable consumption rate as a fraction of total wealth can vary substantially over the interval $(\underline{z}, \bar{z})$ and that optimal behavior can be highly dependent on the risk-return relationship of the durable good and its correlation with the financial market prices.

Despite the relatively elaborate set-up, several extensions are worth considering to make the model more realistic, e.g. introducing a finite time horizon, stochastic interest rates, and a stochastic labor income stream. Until now, such features have only been discussed in models with a perishable consumption good; see, e.g., Bodie, Merton, and Samuelson (1992), Cuoco (1997), and Cocco, Gomes, and Maenhout (1999) for models with stochastic income and Merton (1973) and Campbell and Viceira (1999) for models with utility from perishable consumption in a stochastic interest rate environment. The dynamics of both interest rates and labor income seem to be important factors for individuals' transactions in houses and other major durable goods. Of course, such extensions will also increase the complexity of both the analytical and the numerical analysis. Another direction for future research would be to study the general equilibrium effects of the single agent behavior derived in this paper. In simpler settings, equilibrium effects of durable consumption goods have already been addressed by, e.g., Grossman and Laroque (1990), Caballero (1993), Marshall and Parekh (1999), and Lax (1999).

A Solving the no transaction costs problem

The value function of the no transaction costs problem is defined by

$$\bar{V}(x, p) = \sup_{(\Theta, K, C) \in \bar{\mathcal{A}}(x, p)} \mathbb{E} \left[\int_0^\infty U(C_t, K_t) dt \right],$$

where $U(C, K)$ is given by (2.8), and $\bar{\mathcal{A}}(x, p)$ is as the set $\mathcal{A}(x, k, p)$ defined in Section 2 except that the solvency condition now is $X_t^{\Theta, K, C} \geq 0$ for all $t \geq 0$. The Hamilton-Jacobi-Bellman (HJB) equation corresponding to this problem can be written as

$$\begin{aligned} \rho \bar{V}(x, p) = & \sup_{\substack{\theta \in \mathbb{R}^n, \\ c, k \in \mathbb{R}_+}} \left\{ \frac{1}{1-\gamma} (c^\beta k^{1-\beta})^{1-\gamma} + (r(x - pk) + \theta^\top (\mu - r\mathbf{1}) + (\mu_P - \delta)kp - c) \frac{\partial \bar{V}}{\partial x}(x, p) \right. \\ & + \frac{1}{2} (\theta^\top \sigma \sigma^\top \theta + k^2 p^2 \|\sigma_P\|^2 + 2\theta^\top \sigma \sigma_{P1} kp) \frac{\partial^2 \bar{V}}{\partial x^2}(x, p) + \mu_P p \frac{\partial \bar{V}}{\partial p}(x, p) \\ & \left. + \frac{1}{2} \|\sigma_P\|^2 p^2 \frac{\partial^2 \bar{V}}{\partial p^2}(x, p) + (\theta^\top \sigma \sigma_{P1} + \|\sigma_P\|^2 kp) p \frac{\partial^2 \bar{V}}{\partial x \partial p}(x, p) \right\}. \end{aligned} \quad (\text{A.1})$$

We will show below that under some parameter conditions this highly non-linear partial differential equation has the closed-form solution

$$\bar{V}(x, p) = \frac{1}{1-\gamma} \alpha_v p^{-(1-\beta)(1-\gamma)} x^{1-\gamma},$$

where α_v is a constant. Associated with the solution to the HJB equation are maximizing control values of θ , c , and k , which lead to candidate control policies $\bar{\Theta}_t$, $\bar{C}_t$, and K_t . We also verify that the appropriate transversality condition is satisfied. From the verification theorem for controlled diffusions, see e.g. Theorem IV.5.1 in Fleming and Soner (1993), it follows that $\bar{V}(x, p)$ above is the value function and the candidate control policies are the optimal policies for the no transaction costs case.

A.1 Reducing the dimensionality of the problem

For all $\kappa > 0$, the strategy (Θ, K, C) is admissible with initial wealth x and initial durable price p if and only if the strategy $(\kappa\Theta, K, \kappa C)$ is admissible with initial wealth κx and initial durable price κp . Since $U(\kappa C, K) = \kappa^{\beta(1-\gamma)} U(C, K)$, it follows that $\bar{V}(\kappa x, \kappa p) = \kappa^{\beta(1-\gamma)} \bar{V}(x, p)$. In particular, $\bar{V}(x, p) = p^{\beta(1-\gamma)} \bar{V}(x/p, 1) \equiv p^{\beta(1-\gamma)} \bar{v}(x/p)$. Substituting this into (A.1) and simplifying yield the highly non-linear ordinary differential equation

$$\begin{aligned} (\text{A.2}) \quad & \left(\rho - \beta(1-\gamma) \left[\mu_P - \frac{1}{2} \|\sigma_P\|^2 [1 - \beta(1-\gamma)] \right] \right) \bar{v}(y) \\ = & \sup_{\substack{\hat{\theta} \in \mathbb{R}^n, \\ \hat{c}, k \in \mathbb{R}_+}} \left\{ \frac{1}{1-\gamma} (\hat{c}^\beta k^{1-\beta})^{1-\gamma} + \frac{1}{2} \left(\hat{\theta}^\top \sigma \sigma^\top \hat{\theta} + \|\sigma_P\|^2 (y - k)^2 - 2\hat{\theta}^\top \sigma \sigma_{P1} (y - k) \right) \bar{v}''(y) \right. \\ & \left. + \left((r + [1 - \beta(1-\gamma)] \|\sigma_P\|^2 - \mu_P) (y - k) - \delta k + \hat{\theta}^\top (\mu - r\mathbf{1} - [1 - \beta(1-\gamma)] \sigma \sigma_{P1}) - \hat{c} \right) \bar{v}'(y) \right\} \end{aligned}$$

where we have introduced the scaled control values $\hat{c} = c/p$ and $\hat{\theta} = \theta/p$.

A.2 Solving the reduced HJB equation

We claim that the differential equation (A.2) has the concavely increasing solution

$$(A.3) \quad \bar{v}(y) = \frac{1}{1-\gamma} \alpha_v y^{1-\gamma}$$

with the maximizing control values $\hat{c} = \alpha_c y$, $\hat{\theta} = \alpha_\theta y$, and $k = \alpha_k y$. Substituting (A.3) into (A.2), we get that

$$(A.4) \quad \left(\rho - \beta(1-\gamma) \left[\mu_P - \frac{1}{2} [1 - \beta(1-\gamma)] \|\sigma_P\|^2 \right] \right) \frac{1}{1-\gamma} \alpha_v y^{1-\gamma} \\ = \sup_{\substack{\hat{\theta} \in \mathbb{R}^n, \\ \hat{c}, k \in \mathbb{R}_+}} \left\{ \frac{1}{1-\gamma} (\hat{c}^\beta k^{1-\beta})^{1-\gamma} - \frac{\gamma}{2} \left(\hat{\theta}^\top \sigma \sigma^\top \hat{\theta} + \|\sigma_P\|^2 (y-k)^2 - 2\hat{\theta}^\top \sigma \sigma_{P1} (y-k) \right) \alpha_v y^{-\gamma-1} \right. \\ \left. + \left((r + [1 - \beta(1-\gamma)] \|\sigma_P\|^2 - \mu_P) (y-k) - \delta k + \hat{\theta}^\top (\mu - r\mathbf{1} - [1 - \beta(1-\gamma)] \sigma \sigma_{P1}) - \hat{c} \right) \alpha_v y^{-\gamma} \right\}.$$

Ignoring for now the positivity constraint on $\hat{c}$ and k , the first order conditions for the maximizing control values are

$$(A.5) \quad -\alpha_v y^{-\gamma} + \beta \hat{c}^{\beta(1-\gamma)-1} k^{(1-\beta)(1-\gamma)} = 0,$$

$$(A.6) \quad -\gamma \left(\sigma \sigma^\top \hat{\theta} - \sigma \sigma_{P1} (y-k) \right) + (\mu - r\mathbf{1} - [1 - \beta(1-\gamma)] \sigma \sigma_{P1}) y = 0,$$

and

$$(A.7) \quad \gamma \alpha_v y^{-\gamma-1} \left(\hat{\theta}^\top \sigma \sigma_{P1} - \|\sigma_P\|^2 (y-k) \right) - \alpha_v y^{-\gamma} (r - \mu_P + \delta + [1 - \beta(1-\gamma)] \|\sigma_P\|^2) \\ + (1-\beta) \hat{c}^{\beta(1-\gamma)} k^{-\gamma-\beta(1-\gamma)} = 0.$$

Inserting control values of the form $\hat{c} = \alpha_c y$, $\hat{\theta} = \alpha_\theta y$, and $k = \alpha_k y$, we first get from (A.5) that

$$(A.8) \quad \alpha_v = \beta \alpha_c^{\beta(1-\gamma)-1} \alpha_k^{(1-\beta)(1-\gamma)}$$

and from (A.6) that

$$(A.9) \quad \alpha_\theta = \frac{1}{\gamma} (\sigma \sigma^\top)^{-1} (\mu - r\mathbf{1} - (1-\gamma)(1-\beta) \sigma \sigma_{P1}) - (\sigma^\top)^{-1} \sigma_{P1} \alpha_k.$$

Substituting (A.9) into (A.7) and applying (A.8) yield

$$(A.10) \quad \alpha_c = \frac{\beta}{1-\beta} (r - \mu_P + \delta + (1-\beta)(1-\gamma) \sigma_{P2}^2 + (\mu - r\mathbf{1})^\top (\sigma^\top)^{-1} \sigma_{P1}) \alpha_k + \frac{\gamma\beta}{1-\beta} \sigma_{P2}^2 \alpha_k^2.$$

Substituting the candidate control values back into the equation (A.4) and simplifying yield

$$(A.11) \quad \frac{\rho}{1-\gamma} - \beta \left(\mu_P - \frac{1}{2} [1 - \beta(1-\gamma)] \|\sigma_P\|^2 \right) \\ = \frac{\alpha_c^{\beta(1-\gamma)} \alpha_k^{(1-\beta)(1-\gamma)}}{(1-\gamma) \alpha_v} - \frac{\gamma}{2} (\alpha_\theta^\top \sigma \sigma^\top \alpha_\theta + \|\sigma_P\|^2 (1-\alpha_k)^2 - 2(1-\alpha_k) \alpha_\theta^\top \sigma \sigma_{P1}) \\ + (r + [1 - \beta(1-\gamma)] \|\sigma_P\|^2 - \mu_P) (1-\alpha_k) - \delta \alpha_k + \alpha_\theta^\top (\mu - r\mathbf{1} - [1 - \beta(1-\gamma)] \sigma \sigma_{P1}) - \alpha_c.$$

Applying (A.8), (A.9), and (A.10), Equation (A.11) reduces after tedious manipulations to the quadratic equation

$$(A.12) \quad \Lambda_0 + \Lambda_1 \alpha_k + \Lambda_2 \alpha_k^2 = 0$$

where

$$\begin{aligned} \Lambda_0 &= -\frac{\rho}{\gamma} + \frac{1-\gamma}{\gamma} \left\{ r - (1-\beta)\mu_P + \frac{1}{2}(1-\beta)[1 + (1-\beta)(1-\gamma)] \|\sigma_P\|^2 \right\} \\ &\quad + \frac{1-\gamma}{2\gamma^2} (\mu - r\mathbf{1} - (1-\gamma)(1-\beta)\sigma\sigma_{P1})^\top (\sigma\sigma^\top)^{-1} (\mu - r\mathbf{1} - (1-\gamma)(1-\beta)\sigma\sigma_{P1}), \\ \Lambda_1 &= (1-\gamma)\sigma_{P2}^2 + \frac{1}{1-\beta} (r - \mu_P + \delta + (\mu - r\mathbf{1})^\top (\sigma^\top)^{-1} \sigma_{P1}), \\ \Lambda_2 &= \left(\frac{\gamma}{1-\beta} + \frac{1-\gamma}{2} \right) \sigma_{P2}^2. \end{aligned}$$

For the solution above to satisfy the positivity constraints on $\hat{c}$ and k , the constants α_c and α_k must be positive, in which case the constant α_v will also be positive. We will apply that α_c can be rewritten as

$$(A.13) \quad \alpha_c = \beta\Lambda_1\alpha_k + \beta \left(\Lambda_2 - \frac{1}{2}(1-\gamma)\sigma_{P2}^2 \right) \alpha_k^2$$

and, using (A.12), as

$$(A.14) \quad \alpha_c = -\beta\Lambda_0 - \frac{1}{2}\beta(1-\gamma)\sigma_{P2}^2\alpha_k^2.$$

Consider the quadratic equation (A.12). To determine the number of positive roots, we apply Descartes' rule of sign. Note that Λ_2 is positive (since $\gamma > 0$ and $\beta \in (0, 1)$) and consider the following cases:

1. If Λ_0 and Λ_1 are also positive, there are no positive roots in (A.12).
2. Suppose Λ_0 is positive and Λ_1 is negative. Then there are either zero or two positive roots. In any case, we see from (A.14) that α_c will be negative.
3. If $\Lambda_0 < 0$, there will be a unique positive solution α_k to the quadratic equation (A.12). To ensure a positive α_c , we see from (A.14) that a necessary condition is that $\Lambda_0 < -\frac{1}{2}(1-\gamma)\sigma_{P2}^2\alpha_k^2$. From (A.13) we see that a sufficient condition for $\alpha_c > 0$ is that $\Lambda_1 > 0$.

A.3 The transversality condition

From the first two subsections of this appendix, we can conclude that

$$\bar{V}(x, p) = \frac{1}{1-\gamma} \alpha_v p^{-(1-\beta)(1-\gamma)} x^{1-\gamma}$$

is a solution to the unreduced HJB equation (A.1) if $\Lambda_0 < -\frac{1}{2}(1-\gamma)\sigma_{P2}^2\alpha_k^2$. To ensure that this is indeed the value function, we must verify the transversality condition. The dynamics of the wealth of the investor for any given control strategies (Θ, K, C) is given by (2.5). Substituting in the candidate control

policies $\bar{C}_t = \alpha_c \bar{X}_t$, $\bar{K}_t = \alpha_k \bar{X}_t / P_t$, and $\bar{\Theta}_t = \alpha_\theta \bar{X}_t$, the dynamics of wealth $\bar{X}_t = X_t^{\bar{\Theta}, \bar{K}, \bar{C}}$ becomes a geometric Brownian motion

$$(A.15) \quad \begin{aligned} d\bar{X}_t = & \bar{X}_t \left[r(1 - \alpha_k) + \alpha_\theta^\top (\mu - r\mathbf{1}) + (\mu_P - \delta)\alpha_k - \alpha_c \right] dt \\ & + \bar{X}_t \left(\alpha_\theta^\top \sigma + \alpha_k \sigma_{P1}^\top \right) dw_{1t} + \bar{X}_t \alpha_k \sigma_{P2} dw_{2t}. \end{aligned}$$

In particular, wealth stays strictly positive with probability one. The transversality condition to be shown, cf. Theorem IV.5.1 in Fleming and Soner (1993), is therefore that

$$\lim_{T \rightarrow \infty} e^{-\rho T} \mathbb{E} [\bar{V}(\bar{X}_T, P_T)] = 0,$$

i.e. that

$$(A.16) \quad \lim_{T \rightarrow \infty} e^{-\rho T} \mathbb{E} [\bar{X}_T^{1-\gamma} P_T^{-(1-\beta)(1-\gamma)}] = 0.$$

From (A.15) and (2.2) we get that

$$\mathbb{E} [\bar{X}_T^{1-\gamma} P_T^{-(1-\beta)(1-\gamma)}] = x^{1-\gamma} p^{-(1-\beta)(1-\gamma)} e^{\xi T},$$

where

$$\begin{aligned} \xi = (1 - \gamma) \Big\{ & r(1 - \alpha_k) + \alpha_\theta^\top (\mu - r\mathbf{1}) + (\mu_P - \delta)\alpha_k - \alpha_c - \frac{\gamma}{2} \alpha_\theta^\top \sigma \sigma^\top \alpha_\theta \\ & - \frac{\gamma}{2} \|\sigma_P\|^2 \alpha_k^2 - \gamma \alpha_\theta^\top \sigma \sigma_{P1} \alpha_k + \frac{1}{2} (1 - \beta) [1 + (1 - \beta)(1 - \gamma)] \|\sigma_P\|^2 \\ & - (1 - \beta) \mu_P - (1 - \beta)(1 - \gamma) \alpha_\theta^\top \sigma \sigma_{P1} - (1 - \beta)(1 - \gamma) \alpha_k \|\sigma_P\|^2 \Big\}. \end{aligned}$$

Applying (A.9) and (A.10), this reduces after tedious manipulations to

$$\begin{aligned} \xi &= \rho - (1 - \gamma)(\Lambda_0 + \Lambda_1 \alpha_k + \Lambda_2 \alpha_k^2) + \Lambda_0 + \frac{1}{2} (1 - \gamma) \sigma_{P2}^2 \alpha_k^2 \\ &= \rho + \Lambda_0 + \frac{1}{2} (1 - \gamma) \sigma_{P2}^2 \alpha_k^2, \end{aligned}$$

so that the transversality condition (A.16) is satisfied if and only if

$$\Lambda_0 < -\frac{1}{2} (1 - \gamma) \sigma_{P2}^2 \alpha_k^2,$$

which was already assumed to be the case.

B Proofs of analytical results with transaction costs

B.1 Proof of Theorem 4.1

(a) The upper bound follows from the fact that the maximum expected utility is decreasing in the transaction costs rate λ . The lower bound can be obtained as follows. If $x = \lambda kp$, the only feasible strategy is to liquidate and consequently $V(x, k, p) = 0$ on $\partial\mathcal{S}$. If $(x, k, p) \in \mathcal{S}$, a feasible strategy for the agent is to invest $\eta(x - \lambda kp)$, where $\eta \in (0, 1)$, in the durable good and never trade the durable in the future. In this case, $K_t = \frac{\eta(x - \lambda kp)}{p} e^{-\delta t}$ for all $t \geq 0$. If he, furthermore, invests the remainder in the riskless asset and continuously extracts exactly the added interest rate payments for perishable consumption, i.e. $C_t = (1 - \eta)(x - \lambda kp)r$ for all $t \geq 0$, his life-time expected utility will be

$$\int_0^\infty \frac{1}{1 - \gamma} e^{-\rho t} \left([(1 - \eta)(x - \lambda kp)r]^\beta \left[\frac{\eta(x - \lambda kp)}{p} e^{-\delta t} \right]^{1 - \beta} \right)^{1 - \gamma} dt.$$

This expression is maximized by $\eta = 1 - \beta$, for which the expression gives the alleged lower bound.

(b) Fix (k, p) and consider x_1 and x_2 such that $(x_i, k, p) \in \bar{\mathcal{S}}$, $i = 1, 2$. Denote by (Θ_1, K_1, C_1) and (Θ_2, K_2, C_2) the optimal strategies for $V(x_1, k, p)$ and $V(x_2, k, p)$, respectively. Define for $\eta \in [0, 1]$, $x_\eta = \eta x_1 + (1 - \eta)x_2$, and Θ_η , K_η , and C_η similarly. Because of the linearity of the wealth dynamics in (2.5), we have that

$$X_t^{\Theta_\eta, K_\eta, C_\eta} = \eta X_t^{\Theta_1, K_1, C_1} + (1 - \eta) X_t^{\Theta_2, K_2, C_2} \geq \eta \lambda K_{1t} P_t + (1 - \eta) \lambda K_{2t} P_t = \lambda K_{\eta t} P_t,$$

and thus $(\Theta_\eta, K_\eta, C_\eta) \in \mathcal{A}(x_\eta, k, p)$. Concavity of the value function now follows from the concavity of the utility function, since

$$\begin{aligned} V(x_\eta, k, p) &\geq J^{\Theta_\eta, K_\eta, C_\eta}(x_\eta, k, p) \\ &= \mathbb{E} \left[\int_0^\infty U(C_{\eta t}, K_{\eta t}) dt \right] \\ &\geq \mathbb{E} \left[\int_0^\infty \{ \eta U(C_{1t}, K_{1t}) + (1 - \eta) U(C_{2t}, K_{2t}) \} dt \right] \\ &= \eta J^{\Theta_1, K_1, C_1}(x_1, k, p) + (1 - \eta) J^{\Theta_2, K_2, C_2}(x_2, k, p) \\ &= \eta V(x_1, k, p) + (1 - \eta) V(x_2, k, p). \end{aligned}$$

To show that V is increasing in x , fix (k, p) and pick x_1 and x_2 such that $(x_i, k, p) \in \bar{\mathcal{S}}$ and $x_1 < x_2$. Denote by (Θ_1, K_1, C_1) the optimal strategy corresponding to $V(x_1, k, p)$. Again due to the linearity of the wealth dynamics,

$$X_t^{\Theta_1, K_1, C_1 + (x_2 - x_1)r} = x_2 - x_1 + X_t^{\Theta_1, K_1, C_1},$$

and hence the strategy $(\Theta_t, K_t, C_t) = (\Theta_{1t}, K_{1t}, C_{1t} + (x_2 - x_1)r)$ will belong to $\mathcal{A}(x_2, k, p)$. Therefore,

$$\begin{aligned} V(x_2, k, p) &\geq J^{\Theta_t, K_t, C_t}(x_2, k, p) \\ &= \mathbb{E} \left[\int_0^\infty U(C_{1t} + (x_2 - x_1)r, K_{1t}) dt \right] \\ &> \mathbb{E} \left[\int_0^\infty U(C_{1t}, K_{1t}) dt \right] \\ &= V(x_1, k, p). \end{aligned}$$

(c) By concavity the value function is continuous in the interior of the set $\{x : x \geq \lambda kp\}$.

(d) For all $\kappa > 0$, $(\Theta, K, C) \in \mathcal{A}(x, k, p)$ if and only if $(\kappa\Theta, \kappa K, \kappa C) \in \mathcal{A}(\kappa x, \kappa k, p)$. Also, with the assumed utility function $U(\kappa C, \kappa K) = \kappa^{1-\gamma}U(C, K)$. Therefore, $V(x, k, p)$ is homogeneous of degree $1 - \gamma$ in (x, k) . Similarly, $(\Theta, K, C) \in \mathcal{A}(x, k, p)$ if and only if $(\kappa\Theta, K, \kappa C) \in \mathcal{A}(\kappa x, k, \kappa p)$, and $U(\kappa C, K) = \kappa^{\beta(1-\gamma)}U(C, K)$, which implies that $V(x, k, p)$ is homogeneous of degree $\beta(1 - \gamma)$ in (x, p) .

B.2 Definition of viscosity solutions and proof of Theorem 4.2

We define the viscosity solution concept for the HJB equation (4.10) as follows:

Definition B.1 *An upper semi continuous function $h : S_1 \rightarrow \mathbb{R}$ is a viscosity subsolution of (4.10) on the domain S_1 if for every $\varphi \in C^2(S_1)$ and any local maximum point $z_0 \in S_1$ of the function $h - \varphi$,*

$$(B.1) \quad \max \left\{ H(z_0, h, \varphi', \varphi''), (z_0 - \lambda)^{1-\gamma} \frac{M}{1-\gamma} - h(z_0) \right\} \geq 0.$$

A lower semi continuous function $h : S_1 \rightarrow \mathbb{R}$ is a viscosity supersolution of (4.10) on the domain S_1 if for every $\varphi \in C^2(S_1)$ and any local minimum point $z_0 \in S_1$ of the function $h - \varphi$,

$$(B.2) \quad \max \left\{ H(z_0, h, \varphi', \varphi''), (z_0 - \lambda)^{1-\gamma} \frac{M}{1-\gamma} - h(z_0) \right\} \leq 0.$$

A continuous function $h : S_1 \rightarrow \mathbb{R}$ is a viscosity solution of (4.10) if it is both a subsolution and a supersolution.

For a general overview of viscosity theory, we refer the reader to Crandall, Ishii, and Lions (1992) and Fleming and Soner (1993).

Proof of part (a) of Theorem 4.2. We first prove that v is a viscosity subsolution of (4.10). Let $\varphi \in \mathcal{C}^2[\lambda, \infty)$ and suppose z_0 is a local maximum point of the function $v - \varphi$. We can without loss of generality assume that $v(z_0) = \varphi(z_0)$, and hence $v(z) \leq \varphi(z)$ in some open ball $B(z_0; \eta_0)$ around z_0 with radius η_0 . We assume that there exist strictly positive numbers ε_0 and ε_1 such that

$$(B.3) \quad H(z, v, \varphi', \varphi'') < -\varepsilon_0$$

and

$$(B.4) \quad \frac{M}{1-\gamma} (z_0 - \lambda)^{1-\gamma} - v(z_0) < -\varepsilon_1$$

and seek to obtain a contradiction. Since $\varphi \in \mathcal{C}^2$ and $v(z_0) = \varphi(z_0)$, (B.3) implies that there exists a neighborhood $B(z_0; \eta_1)$ around z_0 such that

$$(B.5) \quad H(z, \varphi, \varphi', \varphi'') < -\varepsilon_0, \quad \forall z \in B(z_0; \eta_1).$$

Furthermore, because v is continuous, (B.4) holds for all z in some neighborhood of z_0 , say $B(z_0; \eta_2)$. Moreover, by the dynamic programming principle, (B.4) implies that it is optimal to refrain from trading the durable asset for $z \in B(z_0; \eta_2)$ and, consequently, the optimal trajectory (Z_t^*) is continuous up to the stopping time $\tau_2 = \inf\{t \geq 0 : Z_t^* \notin B(z_0; \eta_2)\}$. Define the exit time τ_0 by

$$\tau_0 = \inf\{t \geq 0 : Z_t^* \notin B(z_0; \min(\eta_0, \eta_1, \eta_2))\}.$$

Then for $t \in [0, \tau_0]$ we have by assumption that (B.3)-(B.5) hold, and the optimal trajectory Z_t^* is continuous.

Now applying Itô's formula to the process $\{e^{-\bar{\rho}t} P_t^{\beta(1-\gamma)} \varphi(Z_t^*)\}$ and taking expectations yield

$$\mathbb{E} \left[e^{-\bar{\rho}\tau_0} P_{\tau_0}^{\beta(1-\gamma)} \varphi(Z_{\tau_0}^*) \right] = p^{\beta(1-\gamma)} \varphi(z_0) + \mathbb{E} \left[\int_0^{\tau_0} e^{-\bar{\rho}t} P_t^{\beta(1-\gamma)} (\mathcal{L}\varphi(Z_t^*) - \bar{\rho}\varphi(Z_t^*)) dt \right],$$

where the operator $\mathcal{L}$ is given by (4.12) with the optimal strategies inserted for $\hat{\theta}$ and $\hat{c}$. Therefore,

$$\begin{aligned} \mathbb{E} \left[e^{-\bar{\rho}\tau_0} P_{\tau_0}^{\beta(1-\gamma)} \varphi(Z_{\tau_0}^*) \right] &\leq p^{\beta(1-\gamma)} \varphi(z_0) + \mathbb{E} \left[\int_0^{\tau_0} e^{-\bar{\rho}t} P_t^{\beta(1-\gamma)} \left(H(Z_t^*, \varphi, \varphi', \varphi'') - \frac{\hat{C}_t^{\beta(1-\gamma)}}{1-\gamma} \right) dt \right] \\ &< p^{\beta(1-\gamma)} \varphi(z_0) + \mathbb{E} \left[\int_0^{\tau_0} e^{-\bar{\rho}t} P_t^{\beta(1-\gamma)} \left(-\varepsilon_0 - \frac{\hat{C}_t^{\beta(1-\gamma)}}{1-\gamma} \right) dt \right]. \end{aligned}$$

Here, $\kappa \equiv \mathbb{E} \left[\int_0^{\tau_0} e^{-\bar{\rho}t} P_t^{\beta(1-\gamma)} \varepsilon_0 dt \right] > 0$, and therefore, using $v(z_0) = \varphi(z_0)$ and $v(Z_{\tau_0}) \leq \varphi(Z_{\tau_0})$, we get

$$p^{\beta(1-\gamma)} v(z_0) > \mathbb{E} \left[e^{-\bar{\rho}\tau_0} P_{\tau_0}^{\beta(1-\gamma)} v(Z_{\tau_0}^*) + \int_0^{\tau_0} e^{-\bar{\rho}t} P_t^{\beta(1-\gamma)} \frac{\hat{C}_t^{\beta(1-\gamma)}}{1-\gamma} dt \right] + \kappa,$$

which violates the dynamic programming principle, because Z^* was assumed to be the optimal trajectory. Hence, v is a viscosity subsolution of (4.10).

We now prove that v is a viscosity supersolution of (4.10). Let $\varphi \in \mathcal{C}^2[\lambda, \infty)$ and suppose $z_0 \in [\lambda, \infty)$ is a local minimum point of the function $v - \varphi$. Again, we assume without loss of generality that $v(z_0) = \varphi(z_0)$, and note for future reference that $v(z) \geq \varphi(z)$ in some neighborhood of z_0 , say $B(z_0; \eta_0)$. We need to show that

$$(B.6) \quad 0 \geq H(z_0, v, \varphi', \varphi'')$$

and

$$(B.7) \quad 0 \geq \frac{M}{1-\gamma} (z_0 - \lambda)^{1-\gamma} - v(z_0).$$

Since it is always a feasible strategy to rebalance the durable asset, the dynamic programming principle implies that (B.7) is satisfied for all $z \in [\lambda, \infty)$ and thus in particular for z_0 . We now prove that

(B.6) holds. Let $\eta \in (0, \eta_0)$ be such that $B(z_0; \eta) \subset [\lambda, \infty)$. For $c > 0$ and $\theta > 0$ consider a policy $(\hat{C}, \hat{\Theta}) \in \hat{\mathcal{A}}(z_0)$ satisfying $(\hat{C}_t, \hat{\Theta}_t) = (c, \theta)$ for $t \in [0, \tau_\eta]$, where $\tau_\eta = \inf \left\{ t \geq 0 : Z_t^{\hat{C}, \hat{\Theta}} \notin B(z_0; \eta) \right\}$. To simplify notation we shall in the following write Z_t instead of $Z_t^{\hat{C}, \hat{\Theta}}$. Applying Itô's formula to the process $\left\{ e^{-\bar{\rho}t} P_t^{\beta(1-\gamma)} \varphi(Z_t) \right\}$ and taking expectations yield

$$(B.8) \quad p^{\beta(1-\gamma)} \varphi(z_0) = \mathbb{E} \left[e^{-\bar{\rho}\tau_\eta} P_{\tau_\eta}^{\beta(1-\gamma)} \varphi(Z_{\tau_\eta}) - \int_0^{\tau_\eta} e^{-\bar{\rho}t} P_t^{\beta(1-\gamma)} \left(\mathcal{L}^{\hat{C}, \hat{\Theta}} \varphi(Z_t) - \bar{\rho} \varphi(Z_t) \right) dt \right].$$

Furthermore, from the dynamic programming principle we have

$$p^{\beta(1-\gamma)} v(z_0) \geq \mathbb{E} \left[\int_0^{\tau_\eta} e^{-\bar{\rho}t} P_t^{\beta(1-\gamma)} \frac{c^{1-\gamma}}{1-\gamma} dt + e^{-\bar{\rho}\tau_\eta} P_{\tau_\eta}^{\beta(1-\gamma)} v(Z_{\tau_\eta}) \right].$$

Therefore, using (B.8), $v(z_0) = \varphi(z_0)$ and $v(Z_{\tau_\eta}) \geq \varphi(Z_{\tau_\eta})$, we get

$$\begin{aligned} p^{\beta(1-\gamma)} \varphi(z_0) &\geq \mathbb{E} \left[\int_0^{\tau_\eta} e^{-\bar{\rho}t} P_t^{\beta(1-\gamma)} \frac{c^{1-\gamma}}{1-\gamma} dt + e^{-\bar{\rho}\tau_\eta} P_{\tau_\eta}^{\beta(1-\gamma)} \varphi(Z_{\tau_\eta}) \right] \\ &= p^{\beta(1-\gamma)} \varphi(z_0) + \mathbb{E} \left[\int_0^{\tau_\eta} e^{-\bar{\rho}t} P_t^{\beta(1-\gamma)} \left(\mathcal{L}^{\hat{C}, \hat{\Theta}} \varphi(Z_t) - \bar{\rho} \varphi(Z_t) \right) dt + \int_0^{\tau_\eta} e^{-\bar{\rho}t} P_t^{\beta(1-\gamma)} \frac{c^{1-\gamma}}{1-\gamma} dt \right]. \end{aligned}$$

Thus, for all $\eta \in (0, \eta_0)$,

$$0 \geq \mathbb{E} \left[\int_0^{\tau_\eta} e^{-\bar{\rho}t} P_t^{\beta(1-\gamma)} \left(\frac{c^{1-\gamma}}{1-\gamma} + \mathcal{L}^{\hat{C}, \hat{\Theta}} \varphi(Z_t) - \bar{\rho} \varphi(Z_t) \right) dt \right],$$

and consequently

$$(B.9) \quad 0 \geq \min_{z \in B(z_0, \eta)} \left\{ \frac{c^{1-\gamma}}{1-\gamma} + \mathcal{L}^{c, \theta} \varphi(z) - \bar{\rho} \varphi(z) \right\}.$$

In particular, (B.9) holds for $\eta \rightarrow 0$ and thus, by continuity,

$$(B.10) \quad 0 \geq \frac{c^{1-\gamma}}{1-\gamma} + \mathcal{L}^{c, \theta} \varphi(z_0) - \bar{\rho} v(z_0).$$

Finally, maximizing the right hand side of (B.10) over (c, θ) yields (B.6), and hence, v is a viscosity supersolution of (4.10).

Proof of part (b) of Theorem 4.2. We prove that the original value function, $V(x, k, p)$, is continuous at the boundary $\partial \mathcal{S}_1 = \{(x, k, p) \in \mathbb{R}_{++}^3 : x - \lambda kp = 0\}$ of the solvency region. Since $v(z) = V(z, 1, 1)$, it follows that v is continuous in $z = \lambda$. We need the following lemma:

Lemma B.1 *For any given $\lambda > 0$, there exist constants $\eta > 0$, $0 < \kappa \leq \lambda + \eta$, and $\alpha > 0$ such that*

$$V(x, k, p) \leq \psi(x, k, p) \equiv \frac{\alpha}{1-\gamma} (x - \lambda kp)^{1-\gamma}, \quad \forall (x, k, p) \in I_\eta \cup \Gamma_\kappa,$$

where

$$\begin{aligned} I_\eta &= \{(x, k, p) \in \mathbb{R}_{++}^3 : \lambda kp \leq x < (\lambda + \eta)kp\}, \\ \Gamma_\kappa &= \{(x, k, p) \in \mathbb{R}_{++}^3 : kp \leq 1, \lambda kp \leq x < \kappa\}. \end{aligned}$$

To see that Lemma B.1 implies continuity of the value function on $\partial \mathcal{S}_1$, first note that if $(x, k, p) \in \partial \mathcal{S}_1$, the only feasible strategy is to liquidate and, consequently, $V = 0$ on $\partial \mathcal{S}_1$. Secondly, for any boundary

point $(x_0, k_0, p_0) \in \partial S_1$ we have that $0 \leq V(x, k, p) \leq \psi(x, k, p) \rightarrow 0$ for $(x, k, p) \rightarrow (x_0, k_0, p_0)$ through $I_\eta \cup \Gamma_\kappa$.

Proof of Lemma B.1. To simplify notation put $Y_t = (X_t, K_t, P_t)$ and $y = (x, k, p)$. Define the differential operator F , by

$$F(y, \psi, D\psi, D^2\psi) = \sup_{c, \theta} \{A^{c, \theta} \psi(y) + U(c, k) - \rho \psi(y)\},$$

where $D\psi$ and $D^2\psi$ denotes the vector of partial derivatives and the matrix of second order partial derivatives, respectively, and $A^{c, \theta}$ is defined by

$$\begin{aligned} A^{c, \theta} \psi(x, k, p) &= \frac{\partial \psi}{\partial x} (r(x - kp) + \theta^\top (\mu - r\mathbf{1}) + (\mu_P - \delta)kp - c) - \frac{\partial \psi}{\partial k} \delta k + \frac{\partial \psi}{\partial p} \mu_P p + \frac{1}{2} \frac{\partial^2 \psi}{\partial x^2} \|\sigma_P\|^2 p^2 \\ &\quad + \frac{1}{2} \frac{\partial^2 \psi}{\partial x^2} (\theta^\top \sigma \sigma^\top \theta + 2\theta^\top \sigma \sigma_{P1} kp + \|\sigma_P\|^2 k^2 p^2) + \frac{\partial^2 \psi}{\partial x \partial p} (\theta^\top \sigma \sigma_{P1} p + \|\sigma_P\|^2 kp). \end{aligned}$$

Define the auxiliary function $\varphi(z) = \frac{\alpha}{1-\gamma} (z - \lambda)^{1-\gamma}$, where α is a so far arbitrary positive constant. Straightforward computations, utilizing the relation $\psi(x, y, k) = k^{1-\gamma} p^{\beta(1-\gamma)} \varphi(x/(kp))$, imply that

$$(B.11) \quad F(y, \psi, D\psi, D^2\psi) = k^{1-\gamma} p^{\beta(1-\gamma)} H\left(\frac{x}{kp}, \varphi, \varphi', \varphi''\right),$$

where H is defined as in (4.11). Moreover, tedious but straightforward manipulations show that there exist constants B , D , and $\xi > 0$ such that for all $z \geq \lambda$

$$H(z, \varphi, \varphi', \varphi'') = \alpha (z - \lambda)^{1-\gamma} \left(\frac{\gamma}{1-\gamma} \Lambda_0 (z - \lambda)^2 + B(z - \lambda) - \frac{1}{2} \gamma (1 - \lambda)^2 \sigma_{P2}^2 + D(z - \lambda)^\xi \right),$$

where Λ_0 is defined in Section 3. It follows that for any α , constants $\eta > 0$ and $\kappa \in (0, \lambda + \eta]$ exist such that

$$H(x/(kp), \varphi_\eta, \varphi'_\eta, \varphi''_\eta) \leq 0, \quad \forall (x, k, p) \in I_\eta \cup \Gamma_\kappa.$$

Thus by (B.11)

$$(B.12) \quad F(y, \psi, D\psi, D^2\psi) \leq 0, \quad \forall y \in I_\eta \cup \Gamma_\kappa.$$

Define the $\{\mathcal{F}_t\}$ -stopping time $\tau_0 = \inf\{t \geq 0 : Y_t \notin I_\eta \cup \Gamma_\kappa\}$. Itô's formula for semimartingales (see, e.g., Protter (1995, Thm. II.33)) applied to the process $\{e^{-\rho t} \psi(Y_t)\}$ implies

$$(B.13) \quad \mathbb{E}[e^{-\rho \tau_0} \psi(Y_{\tau_0})] = \psi(y) + \mathbb{E} \left[\int_0^{\tau_0} e^{-\rho t} (A^{C, \Theta} \psi(Y_t) - \rho \psi(Y_t)) dt + \sum_{0 \leq t < \tau_0} (\psi(Y_t) - \psi(Y_{t-})) \right].$$

Furthermore, if Y_t jumps at time t , then $X_t = X_{t-} - \lambda K_{t-} P_t$, which implies that $\psi(Y_t) - \psi(Y_{t-}) \leq 0$ for all $t \geq 0$. This latter observation together with (B.12) and (B.13) imply

$$\mathbb{E}[e^{-\rho \tau_0} \psi(Y_{\tau_0})] \leq \psi(y) - \mathbb{E} \left[\int_0^{\tau_0} e^{-\rho t} U(C_t, K_t) dt \right].$$

It can be shown that

$$V(X_{\tau_0}, K_{\tau_0}, P_{\tau_0}) \leq \psi(X_{\tau_0}, K_{\tau_0}, P_{\tau_0})$$

if α is chosen such that $\alpha \geq \alpha_v \left(\frac{\lambda + \eta}{\eta} \right)^{1-\gamma}$. This together with (B.13) imply

$$\mathbb{E} \left[\int_0^{\tau_0} e^{-\rho t} U(C_t, K_t) dt + e^{-\rho \tau_0} V(Y_{\tau_0}) \right] \leq \psi(y).$$

Maximizing the left hand side over all admissible controls $(C, K, \Theta) \in \mathcal{A}(y)$ leads to our claim.

Proof of part (c) of Theorem 4.2. As shown earlier, the function v defined by (4.6) is a viscosity solution of (4.10) with the stated properties. That v is the only such solution follows from the comparison result stated in the following lemma. (Note that an increasing concave function has sublinear growth.)

Lemma B.2 *Suppose that (4.13) is satisfied. Let $\underline{v}$ and $\bar{v}$ be a viscosity subsolution and a viscosity supersolution, respectively, of (4.10) on (λ, ∞) . Furthermore, suppose that both $\underline{v}$ and $\bar{v}$ are continuous and increasing with sublinear growth on $[\lambda, \infty)$ and satisfy $0 = \underline{v}(\lambda) = \bar{v}(\lambda)$ and $\underline{v}(z) \geq \frac{M}{1-\gamma}(z - \lambda)^{1-\gamma}$ for all $z \in [\lambda, \infty)$. Then $\underline{v}(z) \leq \bar{v}(z)$ for all $z \in [\lambda, \infty)$.*

Proof of Lemma B.2. Observe first that the operator H defined in (4.11) can be written as

$$H(z, v, v', v'') = G(z, v', v'') - \hat{\rho}v,$$

where $G : \mathbb{R}^3 \rightarrow \mathbb{R}$ is defined by

$$G(z, a, b) = \sup_{\hat{c} \in \mathbb{R}_+, \hat{\theta} \in \mathbb{R}^n} \left\{ \frac{1}{2} b \left[\hat{\theta}^\top \sigma \sigma^\top \hat{\theta} - 2(z-1) \hat{\theta}^\top \sigma \sigma_{P1} + (z-1)^2 \|\sigma_P\|^2 \right] + \frac{1}{1-\gamma} \hat{c}^{\beta(1-\gamma)} \right. \\ \left. + a \left[(z-1)(r - \mu_P + \delta + [1 - \beta(1-\gamma)] \|\sigma_P\|^2) - \hat{c} + \hat{\theta}^\top (\mu - r\mathbf{1} - [1 - \beta(1-\gamma)] \sigma \sigma_{P1}) \right] \right\}$$

and where

$$\hat{\rho} = \rho - (1-\gamma) \left(\beta \mu_P - \frac{1}{2} \beta [1 - \beta(1-\gamma)] \|\sigma_P\|^2 - \delta \right)$$

by (4.13) is positive.

We assume that

$$(B.14) \quad \sup_{y \in [\lambda, \infty)} \{ \underline{v}(y) - \bar{v}(y) \} > 0$$

and aim to find a contradiction. By assumption, constants $\kappa > 0$ and $\nu \in (0, 1)$ exist such that $\underline{v}(y) \leq \kappa(1+y)^\nu$ for all $y \in [\lambda, \infty)$. Let $\hat{\nu} \in (\nu, 1)$. It follows from (B.14) that for sufficiently small $\varepsilon > 0$ we have

$$(B.15) \quad \sup_{y \in [\lambda, \infty)} \{ \underline{v}(y) - \bar{v}(y) - \varepsilon y^{\hat{\nu}} \} > 0.$$

Moreover, because $\underline{v}$ and $\bar{v}$ are continuous and increasing and $\bar{v} \geq 0$, there exists a number $y^* \in [\lambda, \infty)$ such that

$$\sup_{y \in [\lambda, \infty)} \{ \underline{v}(y) - \bar{v}(y) - \varepsilon y^{\hat{\nu}} \} = \underline{v}(y^*) - \bar{v}(y^*) - \varepsilon (y^*)^{\hat{\nu}}.$$

Note that if $y^* = \lambda$, we have a contradiction to (B.15), since we assumed that $0 = \underline{v}(\lambda) = \bar{v}(\lambda)$.

Define the auxiliary functions $\varphi, \psi : [\lambda, \infty) \times [\lambda, \infty) \rightarrow \mathbb{R}$ by

$$\varphi(y, z) = (\alpha(z - y) - 4\eta)^4 + \varepsilon z^{\hat{\nu}},$$

where $\alpha > 1$, $\eta > 0$, and $\varepsilon > 0$, and

$$\psi(y, z) = \underline{v}(z) - \bar{v}(y) - \varphi(y, z).$$

We shall apply the following results which are demonstrated below:

Claim 1 (a) *The function ψ is bounded on $[\lambda, \infty) \times [\lambda, \infty)$ and its supremum point (y_α, z_α) is attained in the compact set $[\lambda, \bar{\kappa}] \times [\lambda, \bar{\kappa}]$, where $\bar{\kappa} > 0$ is a finite constant independent of α , η , and ε .* (b) *The maximum point (y_α, z_α) has the following properties*

$$(B.16) \quad \lim_{\alpha \uparrow \infty} |z_\alpha - y_\alpha| = 0,$$

$$(B.17) \quad \lim_{\alpha \uparrow \infty} (\alpha(z_\alpha - y_\alpha) - 4\eta)^4 = 0,$$

$$(B.18) \quad \lim_{\varepsilon \downarrow 0} \lim_{\alpha \uparrow \infty} \varepsilon z_\alpha^{\hat{\nu}} = 0.$$

Claim 2 *For all $\alpha > 1$, $\eta > 0$, and $\varepsilon > 0$, there exist numbers $Z < 0$ and $Y > 0$ with $Y + Z < 0$ such that*

$$(B.19) \quad \hat{\rho} \underline{v}(z_\alpha) \leq G(z_\alpha, \frac{\partial \varphi}{\partial z}, Z)$$

$$(B.20) \quad \hat{\rho} \bar{v}(y_\alpha) \geq G(y_\alpha, -\frac{\partial \varphi}{\partial y}, -Y),$$

and¹²

$$(B.21) \quad \begin{pmatrix} Z & 0 \\ 0 & Y \end{pmatrix} \leq \begin{pmatrix} \frac{\partial^2 \varphi}{\partial z^2} & \frac{\partial^2 \varphi}{\partial z \partial y} \\ \frac{\partial^2 \varphi}{\partial z \partial y} & \frac{\partial^2 \varphi}{\partial y^2} \end{pmatrix} = \begin{pmatrix} A + \varepsilon B & -A \\ -A & A \end{pmatrix},$$

with $A = 12\alpha^2(\alpha(z_\alpha - y_\alpha) - 4\eta)^2 \geq 0$ and $B = \hat{\nu}(1 - \hat{\nu})z_\alpha^{\hat{\nu}-2} \leq 0$.

Subtracting (B.20) from (B.19) yields

$$(B.22) \quad \begin{aligned} \hat{\rho}(\underline{v}(z_\alpha) - \bar{v}(y_\alpha)) &\leq G(z_\alpha, \frac{\partial \varphi}{\partial z}, Z) - G(y_\alpha, -\frac{\partial \varphi}{\partial y}, -Y) \\ &= \sup_{\hat{c} \in \mathbb{R}_+, \hat{\theta} \in \mathbb{R}^n} \left\{ \frac{1}{2} Z \left[\hat{\theta}^\top \sigma \sigma^\top \hat{\theta} - 2(z_\alpha - 1) \hat{\theta}^\top \sigma \sigma_{P1} + (z_\alpha - 1)^2 \|\sigma_P\|^2 \right] + \frac{1}{1 - \gamma} \hat{c}^{\beta(1-\gamma)} \right. \\ &\quad \left. + \frac{\partial \varphi}{\partial z} \left[(z_\alpha - 1)(r - \mu_P + \delta + [1 - \beta(1 - \gamma)] \|\sigma_P\|^2) - \hat{c} + \hat{\theta}^\top (\mu - r\mathbf{1} - [1 - \beta(1 - \gamma)] \sigma \sigma_{P1}) \right] \right\} \\ &\quad - \sup_{\bar{c} \in \mathbb{R}_+, \bar{\theta} \in \mathbb{R}^n} \left\{ -\frac{1}{2} Y \left[\bar{\theta}^\top \sigma \sigma^\top \bar{\theta} - 2(y_\alpha - 1) \bar{\theta}^\top \sigma \sigma_{P1} + (y_\alpha - 1)^2 \|\sigma_P\|^2 \right] + \frac{1}{1 - \gamma} \bar{c}^{\beta(1-\gamma)} \right. \\ &\quad \left. - \frac{\partial \varphi}{\partial y} \left[(y_\alpha - 1)(r - \mu_P + \delta + [1 - \beta(1 - \gamma)] \|\sigma_P\|^2) - \bar{c} + \bar{\theta}^\top (\mu - r\mathbf{1} - [1 - \beta(1 - \gamma)] \sigma \sigma_{P1}) \right] \right\}. \end{aligned}$$

¹²For two symmetric n -dimensional matrices C and D , we write $C \geq D$ if $C - D$ is positive definite, that is, if $\xi^\top C \xi \geq \xi^\top D \xi$ for all $\xi \in \mathbb{R}^n$.

Since $Y > 0$ and $Z < 0$, the maximizers in (B.22) are

$$\begin{aligned}\hat{c} &= \left(\frac{4\alpha(\alpha(z_\alpha - y_\alpha) - 4\eta)^3 + \varepsilon \hat{\nu} z_\alpha^{\hat{\nu}-1}}{\beta} \right)^{\frac{1}{\beta(1-\gamma)-1}}, \\ \bar{c} &= \left(\frac{4\alpha(\alpha(z_\alpha - y_\alpha) - 4\eta)^3}{\beta} \right)^{\frac{1}{\beta(1-\gamma)-1}}, \\ \hat{\theta} &= \frac{4\alpha(\alpha(z_\alpha - y_\alpha) - 4\eta)^3 + \varepsilon \hat{\nu} z_\alpha^{\hat{\nu}-1}}{-Z} (\sigma \sigma^\top)^{-1} (\mu - r\mathbf{1} - [1 - \beta(1 - \gamma)]\sigma \sigma_{P1}) + (z_\alpha - 1)(\sigma^\top)^{-1} \sigma_{P1} \\ \bar{\theta} &= \frac{4\alpha(\alpha(z_\alpha - y_\alpha) - 4\eta)^3}{Y} (\sigma \sigma^\top)^{-1} (\mu - r\mathbf{1} - [1 - \beta(1 - \gamma)]\sigma \sigma_{P1}) + (y_\alpha - 1)(\sigma^\top)^{-1} \sigma_{P1}\end{aligned}$$

Tedious, but straightforward, computations yield

$$\begin{aligned}G(z_\alpha, \frac{\partial \varphi}{\partial z}, Z) - G(y_\alpha, -\frac{\partial \varphi}{\partial y}, -Y) &= \left(\frac{1}{1-\gamma} - \beta \right) \left(\hat{c}^{\beta(1-\gamma)} - \bar{c}^{\beta(1-\gamma)} \right) \\ &+ (r - \mu_P + \delta + (\mu - r\mathbf{1})^\top (\sigma^\top)^{-1} \sigma_{P1} + [1 - \beta(1 - \gamma)]\sigma_{P2}^2) \left(4\alpha(z_\alpha - y_\alpha)(\alpha(z_\alpha - y_\alpha) - 4\eta)^3 \right. \\ &+ (z_\alpha - 1)\varepsilon \hat{\nu} z_\alpha^{\hat{\nu}-1} \Big) + \frac{1}{2}\sigma_{P2}^2 (Z(z_\alpha - 1)^2 + Y(y_\alpha - 1)^2) \\ &- \frac{1}{2}(\mu - r\mathbf{1} - [1 - \beta(1 - \gamma)]\sigma \sigma_{P1})^\top (\sigma \sigma^\top)^{-1} (\mu - r\mathbf{1} - [1 - \beta(1 - \gamma)]\sigma \sigma_{P1}) \\ &\times \left(\frac{(4\alpha(\alpha(z_\alpha - y_\alpha) - 4\eta)^3 + \varepsilon \hat{\nu} z_\alpha^{\hat{\nu}-1})^2}{Z} + \frac{(4\alpha(\alpha(z_\alpha - y_\alpha) - 4\eta)^3)^2}{Y} \right)\end{aligned}$$

Claim 3 (a) $\lim_{\varepsilon \downarrow 0} \lim_{\eta \downarrow 0} \lim_{\alpha \uparrow \infty} (\hat{c}^{\beta(1-\gamma)} - \bar{c}^{\beta(1-\gamma)}) = 0$,

(b) $\lim_{\varepsilon \downarrow 0} \lim_{\eta \downarrow 0} \lim_{\alpha \uparrow \infty} (Z(z_\alpha - 1)^2 + Y(y_\alpha - 1)^2) = 0$,

(c) *there exist sequences $(\alpha_n) \uparrow \infty$, $(\eta_n) \downarrow 0$, and $(\varepsilon_n) \downarrow 0$ such that the term*

$$\begin{aligned}& -\frac{1}{2}(\mu - r\mathbf{1} - [1 - \beta(1 - \gamma)]\sigma \sigma_{P1})^\top (\sigma \sigma^\top)^{-1} (\mu - r\mathbf{1} - [1 - \beta(1 - \gamma)]\sigma \sigma_{P1}) \\ & \times \left(\frac{(4\alpha(\alpha(z_\alpha - y_\alpha) - 4\eta)^3 + \varepsilon \hat{\nu} z_\alpha^{\hat{\nu}-1})^2}{Z} + \frac{(4\alpha(\alpha(z_\alpha - y_\alpha) - 4\eta)^3)^2}{Y} \right) \\ & = -\frac{1}{2}(\mu - r\mathbf{1} - [1 - \beta(1 - \gamma)]\sigma \sigma_{P1})^\top (\sigma \sigma^\top)^{-1} (\mu - r\mathbf{1} - [1 - \beta(1 - \gamma)]\sigma \sigma_{P1}) \\ & \times \left(\frac{(\partial \varphi / \partial z)^2}{Z} + \frac{(\partial \varphi / \partial y)^2}{Y} \right)\end{aligned}$$

is non-positive along these sequences.

It follows from Claim 3.c that there exists a sequence $(\alpha_n, \varepsilon_n, \eta_n) \rightarrow (\infty, 0, 0)$ such that

$$\begin{aligned}(B.23) \quad & G(z_\alpha, \frac{\partial \varphi}{\partial z}, Z) - G(y_\alpha, -\frac{\partial \varphi}{\partial y}, -Y) \leq \left(\frac{1}{1-\gamma} - \beta \right) \left(\hat{c}^{\beta(1-\gamma)} - \bar{c}^{\beta(1-\gamma)} \right) \\ & + (r - \mu_P + \delta + (\mu - r\mathbf{1})^\top (\sigma^\top)^{-1} \sigma_{P1} + [1 - \beta(1 - \gamma)]\sigma_{P2}^2) \left(4\alpha(z_\alpha - y_\alpha)(\alpha(z_\alpha - y_\alpha) - 4\eta)^3 \right. \\ & \left. + (z_\alpha - 1)\varepsilon \hat{\nu} z_\alpha^{\hat{\nu}-1} \right) + \frac{1}{2}\sigma_{P2}^2 (Z(z_\alpha - 1)^2 + Y(y_\alpha - 1)^2)\end{aligned}$$

along this sequence. Moreover, from Claim 1, 3.a, and 3.b the right hand side of (B.23) converges to zero, and hence $\psi(y_\alpha, z_\alpha) \leq 0$, along the same sequence. This is a contradiction to (B.15) since

$$\psi(y_\alpha, z_\alpha) > \psi(y^* - 4\eta/\alpha, y^*) = \underline{v}(y^*) - \overline{v}(y^*) - \varepsilon(y^*)^{\hat{\nu}} = \sup_{y \in [\lambda, \infty)} \{ \underline{v}(y) - \overline{v}(y) - \varepsilon y^{\hat{\nu}} \} > 0.$$

It remains to prove Claim 1 to 3.

Proof of Claim 1. (a): Because $\underline{v}$ and $\bar{v}$ have sublinear growth, $\psi(y, z) \rightarrow -\infty$ for $y, z \rightarrow \infty$. Consequently, a constant $\bar{\kappa}$ exists, independent of α , such that

$$\sup_{[\lambda, \infty)^2} \psi(y, z) = \max_{[\lambda, \bar{\kappa}]^2} \psi(y, z).$$

(b): Since $z_\alpha \in [\lambda, \bar{\kappa}]$, (B.18) follows immediately. Moreover, from (a), ψ is bounded. Furthermore,

$$\psi(y_\alpha, z_\alpha) > \psi(y^* - 4\eta/\alpha, y^*) = \underline{v}(y^*) - \bar{v}(y^*) - \varepsilon(y^*)^{\hat{\nu}} = \sup_{y \in [\lambda, \infty)} \{ \underline{v}(y) - \bar{v}(y) - \varepsilon y^{\hat{\nu}} \} > 0,$$

and consequently

$$(\alpha(z_\alpha - y_\alpha) - 4\eta)^4 \leq \underline{v}(z_\alpha) - \bar{v}(y_\alpha) - \varepsilon z_\alpha^{\hat{\nu}} - (\underline{v}(y^*) - \bar{v}(y^*) - \varepsilon(y^*)^{\hat{\nu}}).$$

Since the right hand side is bounded, (B.16) follows. Finally, it can be shown that $y_\alpha \rightarrow y^*$ and $z_\alpha \rightarrow y^*$ as $\alpha \rightarrow \infty$, implying (B.17).

Proof of Claim 2. The existence of numbers Y and Z such that (B.19), (B.20), and (B.21) hold follows from a generalization of Ishii and Lions (1990, Prop. II.3) to the case where the controls are not uniformly bounded, see e.g. Zariphopoulou (1994). To see that $Y > 0$, note from (B.20) that if this is not the case, i.e. if $Y \leq 0$, then

$$\begin{aligned} \hat{\rho}\bar{v}(y_\alpha) &\geq G(y_\alpha, -\frac{\partial\varphi}{\partial y}, -Y) \\ &= \sup_{\bar{c} \in \mathbb{R}_+, \bar{\theta} \in \mathbb{R}^n} \left\{ -\frac{1}{2}Y \left[\bar{\theta}^\top \sigma \sigma^\top \bar{\theta} - 2(y_\alpha - 1)\hat{\theta}^\top \sigma \sigma_{P1} + (y_\alpha - 1)^2 \|\sigma_P\|^2 \right] + \frac{1}{1-\gamma} \bar{c}^{\beta(1-\gamma)} \right. \\ &\quad \left. - \frac{\partial\varphi}{\partial y} [(y_\alpha - 1)(r - \mu_P + \delta + [1 - \beta(1 - \gamma)]\|\sigma_P\|^2) - \bar{c} + \bar{\theta}^\top (\mu - r\mathbf{1} - [1 - \beta(1 - \gamma)]\sigma \sigma_{P1})] \right\} \\ &= \infty, \end{aligned}$$

which contradicts the fact that $\bar{v}(y_\alpha) < \infty$. To see that $Z < 0$, multiply (B.21) with $(1, 1)^\top$ from both sides. This yields $Y + Z \leq \varepsilon B < 0$. As shown above $Y > 0$ and, consequently, it must be the case that $Z < 0$.

Proof of Claim 3. (a) follows from Claim 1.b. To show (b) note that Claim 2 implies that

$$\begin{aligned} Y(y_\alpha - 1)^2 + Z(z_\alpha - 1)^2 &= (z_\alpha - 1, y_\alpha - 1) \begin{pmatrix} Z & 0 \\ 0 & Y \end{pmatrix} \begin{pmatrix} z_\alpha - 1 \\ y_\alpha - 1 \end{pmatrix} \\ &\leq (z_\alpha - 1, y_\alpha - 1) \begin{pmatrix} A + \varepsilon B & -A \\ -A & A \end{pmatrix} \begin{pmatrix} z_\alpha - 1 \\ y_\alpha - 1 \end{pmatrix} \\ &= 12\alpha^2(z_\alpha - y_\alpha)^2(\alpha(z_\alpha - y_\alpha) - 4\eta)^2 + \varepsilon(z_\alpha - 1)^2\hat{\nu}(\hat{\nu} - 1)z_\alpha^{\hat{\nu}-2}. \end{aligned}$$

From Claim 1 this last expression converges to zero for $\alpha \uparrow \infty$, $\eta \downarrow 0$, and $\varepsilon \downarrow 0$. Finally, (c) follows from the fact that for all $\alpha > 1$ we have that

$$\begin{aligned} -\left(\frac{(\partial\varphi/\partial z)^2}{Z} + \frac{(\partial\varphi/\partial y)^2}{Y}\right) &= \frac{Y(\partial\varphi/\partial z)^2 + Z(\partial\varphi/\partial y)^2}{-ZY} \\ &= \frac{Y\left((\partial\varphi/\partial y)^2 - 2\varepsilon\hat{\nu}z^{\hat{\nu}-1}\partial\varphi/\partial y + \varepsilon^2(\hat{\nu}z^{\hat{\nu}-1})^2\right) + Z(\partial\varphi/\partial y)^2}{-ZY} \\ &\rightarrow \frac{(Y+Z)(\partial\varphi/\partial y)^2}{-ZY} \leq 0 \quad \text{for } \varepsilon \downarrow 0, \end{aligned}$$

where the inequality comes from Claim 2.

B.3 Proof of Theorem 4.3

Define the function g as $g(z) = H(z, f, f', f'')$, where $f(z) = (z - \lambda)^{1-\gamma}M/(1-\gamma)$ is the stopping reward function. Furthermore, define the set $G = \{z \in (\lambda, \infty) : g(z) > 0\}$. We first show the following lemma:

Lemma B.3 *The set $G \equiv \{z \in (\lambda, \infty) : g(z) > 0\}$ is either empty or an open interval.*

Proof of Lemma B.3. Tedious, but straightforward, computations yield that g can be written as

$$g(z) = M(z - \lambda)^{-1-\gamma}p(z),$$

where

$$p(z) = \frac{\gamma}{1-\gamma}\Lambda_0(z - \lambda)^2 + B(z - \lambda) - \frac{1}{2}\gamma(1 - \lambda)^2\sigma_{P_2}^2 + D(z - \lambda)^\xi,$$

and the constants B , D , and ξ are given by

$$\begin{aligned} B &= -(1 - \lambda)(1 - \beta)\Lambda_1, \\ D &= \left(\frac{1}{1 - \gamma} - \beta\right)\beta^{1/[1-\beta(1-\gamma)]-1}M^{1/[\beta(1-\gamma)-1]}, \\ \xi &= 1 + \frac{\gamma}{1 - \beta(1 - \gamma)}. \end{aligned}$$

Clearly, $g(z) > 0$ if and only if $p(z) > 0$. Note that $p(\lambda) < 0$, and because $\xi \in (1, 2)$ and $\Lambda_0 < 0$, we have that $p(z) \rightarrow -\infty$ for $z \rightarrow \infty$. A standard computation gives that

$$p'(z) = 2\frac{\gamma}{1-\gamma}\Lambda_0(z - \lambda) + B + \xi D(z - \lambda)^{\xi-1}.$$

We distinguish between the case where $B \geq 0$ and the case where $B < 0$.

If $B \geq 0$, the equation $p'(z) = 0$ has a unique solution $\hat{z}$, which is a maximum point of the function $p(z)$. If $p(\hat{z}) \leq 0$, the set G is empty. If $p(\hat{z}) > 0$, it follows from continuity that G is an interval around $\hat{z}$.

If $B < 0$, the equation $p'(z) = 0$ can have zero, one, or two solutions. If it has no solutions, p is decreasing for all z , and the set G is empty. If it has one solution $\hat{z}$, $p'(z) < 0$ for all z except $z = \hat{z}$, so G is empty. If it has two solutions, $\hat{z}_1$ and $\hat{z}_2$, where $\hat{z}_1 < \hat{z}_2$, it follows that p has a local minimum at $\hat{z}_1$

and a global maximum at $\hat{z}_2$. If $p(\hat{z}_2) \leq 0$, the set G is empty. If $p(\hat{z}_2) > 0$, G is an interval around $\hat{z}_2$.

□

To prove Theorem 4.3, we first show that $G \subseteq \hat{\mathcal{N}}$. To see this, recall that $v(z) \geq f(z)$ for all z . Any z_0 with $v(z) \not\geq f(z)$ is therefore a local minimum point of the function $v - f$ with $v(z_0) - f(z_0) = 0$. Since v by Theorem 4.2 is a viscosity supersolution of the HJB equation, it follows from (B.2) that $H(z_0, v, f', f'') = H(z_0, f, f', f'') = g(z_0)$ is non-positive.

Since we have assumed that G is non-empty, it follows from Lemma B.3 that G is an interval. We want to show that the set $\hat{\mathcal{N}}$ is also an interval. Assuming the contrary, continuity of v implies that an interval (z'_1, z'_2) exists disjoint from G with $v(z'_i) = f(z'_i)$, $i = 1, 2$, and $v(z) - f(z) > 0$ for $z \in (z'_1, z'_2)$. The function $v - f$ will have a maximum on $[z'_1, z'_2]$, say at z' , where of course $v(z') > f(z')$. Since v is a viscosity subsolution of the HJB equation, it follows from (B.1) that

$$H(z', v(z'), f'(z'), f''(z')) \geq 0$$

and since H is strictly decreasing in its second argument by (4.13), we have that

$$g(z) \equiv H(z', f(z'), f'(z'), f''(z')) > 0,$$

which is a contradiction.

Finally, we will demonstrate that $z^* \in (\underline{z}, \bar{z})$. By definition of z^* , we have that

$$f(z^*) = (z^* - \lambda)^{1-\gamma} \frac{M}{1-\gamma} = (z^* - \lambda)^{1-\gamma} (z^*)^{\gamma-1} v(z^*) < v(z^*),$$

which implies that $z^* \in \hat{\mathcal{N}}$.

C Description of the numerical method

We seek to find a solution to the Hamilton-Jacobi-Bellman equation

$$0 = \max \left\{ H(z, v, v', v''), \frac{1}{1-\gamma} (z - \lambda)^{1-\gamma} M - v(z) \right\}, \quad \forall z \in \mathcal{S}_1,$$

where $H(z, v, v', v'')$ is given by the highly non-linear expression

$$\begin{aligned} H(z, v, v', v'') = & \left[\beta(1-\gamma) \left(\mu_P - \frac{1}{2} [1 - \beta(1-\gamma)] \|\sigma_P\|^2 \right) - \bar{p} \right] v(z) + \left(\frac{1}{\beta(1-\gamma)} - 1 \right) \beta^{\frac{1}{1-\beta(1-\gamma)}} v'(z)^{\frac{\beta(1-\gamma)}{\beta(1-\gamma)-1}} \\ & + (r - \mu_P + \delta + (\mu - r\mathbf{1})^\top (\sigma^\top)^{-1} \sigma_{P1} + [1 - \beta(1-\gamma)] \sigma_{P2}^2) (z - 1) v'(z) + \frac{1}{2} \sigma_{P2}^2 (z - 1)^2 v''(z) \\ & - \frac{1}{2} (\mu - r\mathbf{1} - [1 - \beta(1-\gamma)] \sigma \sigma_{P1})^\top (\sigma \sigma^\top)^{-1} (\mu - r\mathbf{1} - [1 - \beta(1-\gamma)] \sigma \sigma_{P1}) \frac{v'(z)^2}{v''(z)}, \end{aligned}$$

which is obtained by substituting the optimal scaled controls (4.14)-(4.15) into (4.11).

The solution $v(z)$ is a solution to a free boundary equation on the continuation region $(\underline{z}, \bar{z})$ such that $v(z) \geq \frac{1}{1-\gamma} (z - \lambda)^{1-\gamma} M$, and with the following value matching and “smooth pasting” conditions on the boundaries

$$(C.1) \quad v(\underline{z}) = \frac{1}{1-\gamma} M (\underline{z} - \lambda)^{1-\gamma}, \quad v'_+(\underline{z}) = M (\underline{z} - \lambda)^{-\gamma},$$

$$(C.2) \quad v(\bar{z}) = \frac{1}{1-\gamma} M (\bar{z} - \lambda)^{1-\gamma}, \quad v'_-(\bar{z}) = M (\bar{z} - \lambda)^{-\gamma}.$$

Here, v'_+ and v'_- denote the right and left derivatives, respectively. A solution to this system is given by $(M, \underline{z}, \bar{z})$ so that the equations above are satisfied, and such that M solves the equation

$$M = (1 - \gamma) \sup_{z \geq \lambda} z^{\gamma-1} v(z).$$

As Grossman and Laroque (1990) we apply a stepwise numerical procedure to find the optimal values $(M, \underline{z}, \bar{z})$:

- (a) Guess $M = M_0$.
- (b) Solve the free boundary problem, with $M = M_0$, as follows:
 - (i) Guess $\underline{z} = \underline{z}_0$.
 - (ii) Solve the ODE $H(z, v, v', v'') = 0$ using (C.1) as initial conditions until the value matching part of equation (C.2) is satisfied.
 - (iii) If the smooth pasting condition in (C.2) is satisfied, a candidate value function $v_{M_0}(z)$ is found, otherwise repeat steps (i) and (ii).
- (c) Compute the implied $M_0^* = (1 - \gamma) \sup_{z \in [\underline{z}_0, \bar{z}_0]} z^{\gamma-1} v_{M_0}(z)$. If $M_0^* = M_0$ the problem is solved, otherwise go to (a).

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